Measuring Habit Formation*

Jiannan Zhou

July 9, 2026

Abstract

For millennia, habit formation has been recognized as essential across disciplines, yet insufficient economic measurement has fueled controversies over its existence, parameterization, and implications. We specify total-consumption variations that separately identify habit's internal and external formation, decay speed, and welfare impacts relative to consumption and peer effects. Implementing them in survey experiments, we find that habit forms internally, decays by two-thirds annually, and exerts a welfare impact equal to 63% of consumption's and three times that of peer effects. Contrary to common belief, neither habit formation nor peer effects alone can resolve the Easterlin paradox, but their combination largely can.

Keywords: Habit formation, consumption, micro evidence, Easterlin paradox, measurement
JEL codes: D15, D91, C83, E21, G12, I31

*Zhou: School of Economics, Shandong University (jiannan.zhou@icloud.com). I am indebted to Miles Kimball, Alessandro Peri, Martin Boileau, and Xingtan Zhang for their advice and support during this paper's development. I thank Daniel Benjamin, Andrew Caplin, Michael Woodford, John Cochrane, Kelsey O'Connor, Adam Szeidl, Colin Camerer, Marciano Siniscalchi, Tong Wang, Jian Li, Claudia Sahm, Hanming Fang, Richard Mansfield, Carlos Martins-Filho, Adam McCloskey, Matthew Rognlie, Sanjai Bhagat, Sergey Nigai, Scott Savage, Terra McKinnish, Vincent Mastantuno, Tony Cookson, Nathalie Moyen, Scott Schuh, Edward Van Wesep, Jin-Hyuk Kim, Nicholas Flores, Charles de Bartolomé, Brendan Epstein, and Shuang Zhang, and many seminar and conference participants for their helpful comments. Financial support from Miles Kimball is gratefully acknowledged. This work utilized the Summit supercomputer and the Alpine high-performance computing resource at the University of Colorado Boulder. The research was approved by the ethics committees at Shandong University in 2024 and the University of Colorado Boulder in 2018 (Protocol No. 180012). The author has no material financial interests that relate to the research described in this paper.

"The difficulty [in composing *The General Theory*] lies, not in the new ideas,
but in escaping from the old ones[—'habitual modes of thought and expression'],
which ramify, for those brought up as most of us have been, into every corner of our minds."
—John Maynard Keynes, 1936

1 Introduction

For more than two thousand years, habit formation has been recognized as a fundamental determinant of human behavior. Aristotle observed that "moral virtue comes about as a result of habit" (*Nicomachean Ethics* 2.1), and Confucius that "what nature put together, habit separates" (*The Analects* 17.2). In economics, prominent figures such as Adam Smith and John Maynard Keynes highlighted its profound impact on character and consumption. In *The Wealth of Nations* (1776), Smith observed that "the difference between the most dissimilar characters, between a philosopher and a common street porter, seems to arise not so much from nature, as from habit, custom, and education." Keynes argued in *The General Theory* (1936) that "a man's habitual standard of life usually has the first claim on his income . . . when employment falls to a low level, aggregate consumption will decline by a smaller amount than that by which real income has declined, by reason both of the habitual behavior of individuals and also of the probable policy of governments." Habit formation has since been incorporated into modern economic theory, with early contributions by Duesenberry (1949), Pollak (1970), and Spinnewyn (1981) in microeconomics; Ryder and Heal (1973), Carroll et al. (2000), and Fuhrer (2000) in macroeconomics; and Abel (1990), Constantinides (1990), and Campbell and Cochrane (1999) in finance.

Despite its enduring significance across time and disciplines and its theoretical prominence in economics, empirical work has struggled to deliver clear and consistent micro-level evidence. As a result, economists have yet to reach consensus on the most basic features of habit formation: whether consumption habit exists (Cochrane, 2017), how fast it decays, and how much it matters for welfare. Moreover, the limited evidence on the parameters of habit formation models raises overfitting concerns that undermine model credibility (Breeden et al., 2015) and the further development of habit-based models, as micro-level evidence serves as a foundation for predictions and policymaking (Stiglitz, 2018; Lucas, 1976).

The obstacle is measurement. Economic theories of habit formation center on how past total consumption affects current utility through habit, yet neither habit nor utility is directly observed. Furthermore, observed intertemporal choices are confounded by income, wealth, expectations, and other constraints that are often themselves unobservable. Existing measurements are correspondingly limited: they focus on habit's existence and intensity while largely neglecting its decay speed and welfare impacts, and they often infer from observational data under tenuous identifying assumptions, such as restrictive parametric structures or partial-spending proxies for total spending. Laboratory and field experiments measure habit in behaviors such as exercise and energy use, but neither can feasibly generate the total-spending variations over extended periods that a full assessment of habit formation's existence, persistence, and welfare impacts

requires. Consequently, despite three decades of research, economic measurements of total-consumption habit formation remain limited and inconsistent, with substantial variation across contexts.

This paper introduces an approach that can directly measure key aspects of the general habit-formation functional $u(C, H)$, circumventing the identification challenges in existing methods. Our focus throughout is *total-consumption* habit formation—the form used in standard models of intertemporal consumption choice—as distinct from habits in specific goods or cue-triggered behaviors; consistent with the literature, we hereafter refer to it simply as habit formation.[1] We measure habit's existence by assessing whether habit, H, defined as a function of past total consumption (C), reduces utility ($u_H < 0$)—the standard sign convention for habit formation in economics, as opposed to $u_H > 0$ (such as durability) or $u_H = 0$ (time-separability).[2] We also measure habit's decay speed by gauging the speed at which H changes over time, and habit's welfare impacts by measuring how H affects u. Under minimal restrictions on $u(C, H)$—monotonicity in C ($u_C > 0$), differentiability, and exponential time discounting—we derive separate identification of these parameters through total consumption variations, the type of variation most directly informative about total-consumption habit formation. In principle, the identifying variations are implementable through any approach that can isolate them. We implement them through hypothetical choice survey experiments to circumvent the restrictive assumptions often required to isolate habit formation through observational data and feasibly achieve identification that remains elusive in laboratory and field experiments, while broadening the scope of evidence on habit formation in the literature.

Hypothetical choices are well-grounded in economics (see, e.g., Barsky et al., 1997; Manski, 2004; Ameriks et al., 2020; Stantcheva, 2023) and, when carefully designed, align closely with real-world decisions (Hainmueller et al., 2015; Wiswall and Zafar, 2018; Matousek et al., 2022; Kumar et al., 2023), particularly in spending contexts (Coibion et al., 2024). They do raise validity concerns (e.g., Diamond and Hausman, 1994), which we address in Section 2.3. Unlike traditional real-world data, which often suffer from identification issues, our implementation follows the "data engineering" strategy—generating new forms of data closely linked to model constructs (Caplin, 2025, 2024; Almås et al., 2024)—which has been applied to expectations (Roth and Wohlfart, 2020; Coibion et al., 2022), earnings (Weidmann and Deming, 2021), consumer spending (Ameriks et al., 2007; Fuster et al., 2021; Coibion et al., 2024),

[1] Habits for specific goods map into deep-habit models, where habit forms at the level of individual goods or varieties (Ravn et al., 2006). Cue-triggered behaviors align with cue- or automation-based models of repeated choice within specific environments (Matyskova et al., 2020; Camerer et al., 2024). These models are important but do not directly discipline the total-consumption functional studied here (Webb et al., 2026).

[2] Our identification operates at this functional level: "habit formation exists" is a claim about the sign of u_H, not about a unique underlying mechanism. This functional interpretation is consistent with habituation, the reduced response to repeated stimulation documented across species (Liu et al., 2025; Pinsker et al., 1970), but our design does not separately identify habituation from other backward-looking mechanisms that generate the same functional representation, such as backward-looking reference dependence (see Section 2.1). Other alternatives, including expectations-based reference dependence and preferences over consumption sequences, carry different implications for response patterns the survey reports; Appendix F.7 examines them. The cross-partial derivative $u_{CH} > 0$, often associated with habit formation in the literature (e.g., Constantinides, 1990), is implied by $u_H < 0$ under canonical habit specifications with standard parameter values. We defer measurement of such second-order derivatives to a companion paper.

and firm decisions (Kumar et al., 2023).

Parameters elicited from hypothetical questions have in turn been carried into quantitative work. Ameriks et al. (2020) combine strategic survey questions on long-term care and bequests with administrative wealth data to estimate a structural model of late-in-life saving. The hypothetical income gambles of Barsky et al. (1997), fielded in the Health and Retirement Study, were subsequently adopted by the Panel Study of Income Dynamics, and the risk tolerance imputed from them is now widely used in empirical work (Kimball et al., 2008, 2009). In macroeconomics, hypothetical windfall questions are one of the three recognized approaches used to measure marginal propensities to consume (Jappelli and Pistaferri, 2014; Fuster et al., 2021), whose reproduction is central to the usefulness of heterogeneous-agent models (Kaplan and Violante, 2022). In the present setting, hypothetical choices offer two advantages: they can identify habit parameters that remain elusive under traditional methods, and they provide complementary evidence for parameters traditional methods can also address, since hypothetical bias tends to be orthogonal to the identification challenges that limit traditional methods. To strengthen validity, we anchor spending variations in empirical realities, use a two-wave design to mitigate response error, and add experimental controls that isolate habit formation from confounds such as durable goods. Our findings are corroborated by respondents' own explanations and by related behaviors reported in the literature, demonstrating convergent validity. They further exhibit predictive validity: respondents who reported faster habit adjustment in our hypothetical scenarios showed faster hedonic adaptation to the 2024 U.S. presidential election outcome, as measured in an independent survey fielded two months later than ours.

Using this approach, we document new micro evidence on habit formation, covering its existence, decay speed, and welfare impacts. First, habit forms internally, based on an individual's own past consumption; we find no evidence that it also forms externally, based on others' past consumption. This finding supplies direct evidence for the preference-based formulation of habit formation, and specifically for its internal-habit variants. Second, habit decays at a rate of two-thirds per year, a speed that could significantly impact the performance of habit formation models. Third, habit formation has a substantial welfare impact, equivalent to 63% of consumption's impact and three times that of peer effects. Finally, contrary to common belief, habit formation alone—or peer effects alone—cannot explain the Easterlin paradox, but their combination provides a plausible explanation.

To obtain these results, we use a general model that is agnostic about whether habit formation exists and how it enters utility, while nesting standard models with habit formation. This generality matters because rejecting a specialized version does not refute the general idea of habit formation (Cochrane, 2017). We then specify preference parameters governing the existence, decay speed, and welfare implications of habit formation (Section 2.1), and identify them through total-consumption variations implemented in a survey on Prolific, yielding a broadly representative U.S. sample of about 2,100 responses.

Following detailed instructions and comprehension checks, the survey presented respondents with a series of binary choice questions asking about the (utility) ranking of spending profiles. The profiles

differ in the timing and level of spending, with both dimensions anchored to ensure familiarity. Such hypothetical variations are routinely used to identify preference parameters in the literature (e.g., Barsky et al., 1997; Ameriks et al., 2020). Leveraging humans' natural capacity for counterfactual thinking about both past and future variations (Epstude and Roese, 2008), the differing spending profiles induce variations in habitual spending levels, thereby influencing the utilities of the profiles. The ranking of the utilities is thus a function of the spending difference and the preference parameters. Given the spending difference, inverting the utility ranking indicated in the survey responses recovers the preference parameters. Section 2.2 details the design of our survey experiments, such as the visual presentation of the spending profiles and the hypothetical scenario for reducing cognitive load and controlling for confounding factors. Section 2.3 discusses our strategies to enhance the validity of our hypothetical choices and presents evidence of their effectiveness. Section 2.4 summarizes the sample. Section 2.5 presents the statistical model for extracting reliable responses from the two survey waves while controlling for response errors not addressed by the survey experimental design.

Building on this framework, Section 3 details the measurement of the preference parameters of interest, starting with the existence of habit formation (Section 3.1). Because internal and external habit formation can have different implications for optimal tax policy and welfare analysis (Carroll et al., 1997; Ljungqvist and Uhlig, 2000), we measure them separately. To identify internal habit formation, we vary own past spending while holding future spending fixed. Choosing the profile with the lower past spending for a more satisfying future experience implies $u_H < 0$ and therefore internal habit formation. About two-thirds of respondents made this choice in each wave, and estimation accounting for response errors and respondents' open-ended explanations both confirm internal habit formation. A parallel question on external habit formation, varying only other people's past spending, provides no evidence supporting its existence.

Given the existence of habit formation, we next measure the speed of habit decay (Section 3.2), which is central to habit-based models but has received far less empirical attention than habit intensity. Because the decay rate determines how quickly the effect of past consumption dissipates, we identify it by varying the persistence of past spending: if sustained past spending affects utility less than recent transitory past spending, habit decays faster than a threshold (Proposition 1). This identification parallels the choice-based measure of habit decay proposed by Rozen (2010). The estimated decay speed is about two-thirds per year, with a confidence interval ([0.58, 0.69] at the 95% level) that is narrow relative to the range used in habit-based models—from full decay after a quarter in many macroeconomic applications (e.g., Smets and Wouters, 2007; Christiano et al., 2005) to about 11% per year in Campbell and Cochrane (1999). Appendix E illustrates the quantitative importance of the decay speed in a widely cited habit formation framework.

Section 3.3 details the measurement of two preference parameters that govern the welfare impacts of habit formation. In particular, the slope of the indifference curve, $-\frac{u_H}{u_C}$, measures the welfare impact of habit formation relative to that of one's own consumption. The ratio $\frac{u_{C_{\text{others}}}}{u_H}$ assesses the relative welfare impact of habit formation and peer effects from others' contemporaneous consumption (C_{others}). Both

parameters offer insights into the resolution of the Easterlin paradox. As ratios, the identification of these parameters naturally depends on trade-offs: habit versus consumption (for $-\frac{u_H}{u_C}$), and peer effects versus habit (for $\frac{u_{C_{\text{others}}}}{u_H}$). Leveraging these tailored identifying variations, our findings imply a habit intensity of approximately 0.6, providing direct micro-level evidence for a value often used in model calibrations. Furthermore, habit formation's welfare effect is found to be three times that of peer effects.[3]

Section 4 investigates the potential of habit formation to explain the half-century-old Easterlin paradox: happiness and income are positively correlated in the cross-section and short run but essentially uncorrelated in the long run (Easterlin, 1973, 1974; Stevenson and Wolfers, 2008; Easterlin, 2017; Easterlin and O'Connor, 2025). Although habit formation has been proposed as an explanation (Easterlin, 1995; Clark et al., 2008), direct evidence on whether it can actually account for the paradox has been absent. Using this paper's measurements, we find that neither habit formation nor peer effects—another popular explanation—alone can fully account for the happiness–income dynamics. Combining the two, however, can account for much of the observed pattern across all three dimensions: cross-section, short run, and long run. Individuals eventually habituate to income and observe others' similar income adjustments, explaining the negligible long-run happiness–income gradient. Before full habituation or awareness of others' income changes, happiness tracks income, driving the positive short-run gradient. Wealthier individuals and countries accumulate more short-term happiness, accounting for the positive cross-sectional gradient. The mechanism implies that, despite the negligible long-run happiness–income gradient, sustained economic growth remains important for maintaining the stabilized level of happiness.

Section 5 concludes.

Related literature. The primary contribution of this paper is a methodology for directly measuring total-consumption habit formation. It adds to four literatures: micro evidence on habit formation, the Easterlin paradox, reference dependence, and structural preference elicitation.

Over the past three decades, the study of micro evidence on habit formation has become an active field. Using mostly observational data, it centers on the existence of internal habit formation and yields divergent findings: Dynan (2000), Meghir and Weber (1996), and Brunnermeier and Nagel (2008) find no evidence, while Crawford (2010), Naik and Moore (1996), and Khanal et al. (2018) find support, and Browning and Collado (2007) report mixed results. The limited evidence on external habit formation is likewise conflicting, as Dynan and Ravina (2007) support its presence while Brunnermeier and Nagel (2008) do not. Moreover, habit intensity estimated from micro data often falls short of the levels required to account for macroeconomic data (see Havranek et al., 2017, for a review). Relative to this body of evidence, the parameters measured here fall into three groups: the habit decay rate and the relative welfare impact of habit formation and peer effects, for which direct micro estimates are essentially absent; the

[3]Peer effects and external habit formation share the comparison with others' consumption, contemporaneous in the first case and lagged in the second; the evidence in this paper supports the former but not the latter.

existence of internal and of external habit formation, on which the existing evidence conflicts; and habit intensity, on which micro and macro evidence disagree.

Experimental studies more consistently find evidence of habit formation, though often in non-consumption settings—gym attendance and handwashing (Charness and Gneezy, 2009; Royer et al., 2015; Hussam et al., 2022) and state-recognition tasks (Matyskova et al., 2020). Allcott and Rogers (2014) find that the effects of repeated home energy reports on household energy use decay at 10–20% per year once the reports stop. Allcott et al. (2022) estimate a structural habit model from a randomized experiment on social media use, and report that their data cannot separately identify the direct effect of habit on utility—a parameter this paper measures in both sign and relative magnitude. Camerer et al. (2024) embed a neurally grounded habit process in a structural model and test it on tuna purchase and social-media data, measuring the automaticity with which particular choices are repeated. The object measured here is different: the total-consumption functional $u(C, H)$ that habit formation models take as primitive, which the automaticity of specific-good choices does not directly discipline (Webb et al., 2026). Hypothetical intertemporal choices in Kapteyn and Teppa (2003) favor a multiplicative habit model over a non-habit alternative.

This paper contributes to the literature by developing identifying variations that can directly measure total-consumption habit formation. Rozen (2010) and Tserenjigmid (2020) axiomatize linear habit formation, in which utility depends on consumption net of a reference point formed linearly from past consumption, and Rozen (2010) shows that the reference point's weights on past consumption can be recovered from choices over consumption streams without knowledge of the felicity function or the discount factor. Our identification of the decay rate, which governs how these weights decline with time, shares this independence. Our approach differs in three respects: our rankings compare consumption streams across consumption histories, a comparison included in the primitive of Tserenjigmid (2020), whereas Rozen's choices are made under a given history; we test, rather than impose, the sign of habit's effect on utility; and we extend the identification to external habit and peer effects.

Using the recently termed "data engineering approach" (Caplin, 2025, 2024; Almås et al., 2024), our survey implementation of the identifying variations isolates habit formation from confounding factors in observational data and avoids the prohibitive costs of studying total-consumption habit formation in laboratory and field settings, addressing concerns due to heterogeneity of habit formation across behaviors (Wood and Rünger, 2016; Buyalskaya et al., 2023). The validity of these identifying variations rests on standard economic assumptions, independent of the survey implementation. This approach yields direct micro-level evidence on the existence and intensity of habit formation, while addressing concerns in the literature, such as the reliance on partial spending data, parametric assumptions, and model fit over direct examination of habit's effect on preference structure. By tailoring identifying variation for each of the habit-related parameters, this paper broadens the scope of micro evidence on habit formation with fewer constraints from specification and data availability: it provides direct micro estimates of the habit decay rate and of the relative welfare impact of habit formation and peer effects, which have been essentially absent.

A more focused literature tests habit formation models while relaxing parametric assumptions: Chen and Ludvigson (2009) let habit evolve nonparametrically under additive habit and power utility specifications. Crawford (2010) relaxes parametric assumptions on both utility and habit evolution. We go further with a nonparametric utility function that relaxes joint concavity in consumption and habit, thereby encompassing the common multiplicative habit specification. We further accommodate infinitely many lags in habit evolution and leverage variation directly informative of total-consumption habit formation.[4]

In the five decades since the Easterlin paradox was introduced (Easterlin, 1973, 1974), many explanations have been proposed to resolve it, but how to validate them—especially those based on unobservable constructs such as habit formation (Easterlin, 1995; Clark et al., 2008)—remains an open question. This paper offers a perspective for tackling the issue: measuring the constructs with minimal restrictions and conducting calibrated quantitative investigations to assess the extent to which each explanation can contribute to resolving the paradox. Using this approach, we corroborate habit formation's key role in explaining the paradox, yet challenge the view that it alone suffices. This study further advances the literature by controlling for and assessing changes in reporting scales à la Benjamin et al. (2023), thereby differentiating the explanation through habit formation (and peer effects) from the rescaling explanation (Deaton, 2008; Prati and Senik, 2024). The paradox's near-zero long-run happiness–income gradient has been used to critique growth-centric policies (e.g., Alexander, 2012); our explanation implies instead that sustained economic growth, though perhaps insufficient to elevate long-run happiness, is necessary to maintain it.

This paper also relates to the extensive literature on reference dependence (Markowitz, 1952; Kahneman and Tversky, 1979; Tversky and Kahneman, 1991), where reference points are typically anchored to the status quo (e.g., Masatlioglu and Ok, 2014) or expectations (e.g., Kőszegi and Rabin, 2006). This literature has linked habit formation with reference dependence, leading to the introduction of backward-looking reference points (e.g., Bowman et al., 1999; Post et al., 2008). DellaVigna et al. (2017) and Thakral and Tô (2021) provide evidence for such reference points in the labor supply domain, where reference points depend on past income or recent earnings; the corresponding evidence in the consumption domain, as discussed above, often relies on partial spending and has been mixed. We provide direct evidence for backward-looking reference points in the total-consumption setting, and quantify their evolution and welfare impact.

Finally, the paper contributes to structural preference elicitation. This literature has mostly focused on measuring risk aversion, time discounting, intertemporal substitution, and well-being trade-offs under fully parametric preferences (e.g., Barsky et al., 1997; Kimball et al., 2008, 2009, 2024b) or first-order approximations for semiparametric and nonparametric preferences (e.g., Benjamin et al., 2014b, 2019). We extend it to habit formation: by designing variations in past consumption, we identify habit decay speed exactly and its welfare impacts through first- and second-order approximations under a nonparametric utility function.

[4] In order to measure habit's decay rate and welfare impacts, we specialize to a standard linear habit evolution (Section 2.1). For the existence of habit formation, we impose no restriction on the habit evolution beyond that past spending raises habit.

2 Methodology

2.1 Model and Preference Parameters

To address the gaps in the existing evidence, we use a general model that is agnostic regarding the existence and specification of habit formation, while retaining standard assumptions needed to speak to existing models. Specifically, the agent maximizes

$$\int_0^\infty e^{-\rho t} u\left(C_t, H_t\right) dt,$$

where C_t is the agent's total spending, H_t is their habitual level of spending (henceforth, habit), and ρ is the time discount rate. Unless otherwise specified, the time index is omitted hereafter for brevity. We follow the literature in maintaining exponential time discounting and the restriction that past consumption influences current utility, if at all, only through the habit H. The nonparametric utility function u is assumed three times continuously differentiable and strictly increasing in consumption ($u_C > 0$). Beyond these assumptions, the functional form of $u(C, H)$ is left unrestricted; in particular, the analysis imposes no parametric form on whether and how habit enters utility. The utility function may depend on additional variables, such as leisure. Since these variables remain constant in the identifying variations, omitting them from the explicit utility function entails no loss of generality. When examining changes in variables beyond own consumption and habit—such as others' current consumption—these additional variables will be explicitly incorporated into the utility function.

For the existence of habit formation, we do not need to specify a habit evolution equation, except to require that habit rises with past consumption, a monotonicity shared by effectively all existing habit evolution equations. Quantifying habit's decay rate and welfare impacts, however, requires putting habit on a scale. We therefore assume habit evolves according to

$$\dot{H} = \theta\left(C - H\right), \tag{1}$$

where θ is the habit decay rate. This equation makes habit an exponentially weighted average of past spending, with the weight on each period's spending declining with the elapsed time at rate θ.

We use this specification for four reasons. First, it is the most widely used habit evolution equation in the literature, with the discrete-time expression $H_t = \left(1 - e^{-\theta}\right) C_t + e^{-\theta} H_{t-1}$. Researchers have used different habit evolution equations; however, their differences either amount to a simple rescaling of the unit of habit (e.g., the habit evolution equation of Constantinides, 1990 and the autoregressive habit stock axiomatized by Rozen, 2010 and Tserenjigmid, 2020) or vanish in the steady state (e.g., the habit evolution equation of Campbell and Cochrane, 1999).[5] Second, the habit evolution equation can

[5] In the autoregressive model of Rozen (2010) and Tserenjigmid (2020), period utility is $u(C - sH)$ in our notation, where s is the sum of the habit coefficients, the weights on past consumption in the reference point. Habit is responsive to weaning—it can be offset by a finite total amount of compensating consumption—if $s < 1$, and persistent if $s = 1$ (Rozen, 2010). Our normaliza-

accommodate mixtures of internal and external habits by replacing C with the relevant weighted average of own and others' consumption (e.g., Grishchenko, 2010), with negligible impact on the results presented in this paper. Third, this habit evolution equation provides an intuitive unit for habit, identical to that of consumption, consistent with H's definition as the habitual level of spending. For example, a person who has spent \$5,000 per month for as long as they can remember has a habitual spending level of \$5,000 per month. More generally, with habit measured in this unit, smooth nonlinear habit evolution equations have equation (1) as their first-order expansion around steady state.[6] Fourth, although consumption may itself be chosen with future habit in mind, as in rational-addiction models (Becker and Murphy, 1988), the equation describes how habit responds to consumption rather than how consumption is chosen, and this mechanical evolution is supported by extensive evidence from the biology literature showing habituation across diverse species, including single-celled organisms without neural systems (Boisseau et al., 2016).

These assumptions complete the model's specification. The model is intentionally parsimonious, allowing it to nest more specialized models whose extra structures are held constant in the experiments. Furthermore, we do not explicitly model budgetary constraints, as the respondents choose among complete, budget-feasible spending profiles (see Section 2.2 for details).

Our identification operates at the level of the functional $u(C, H)$ and is agnostic about the underlying mechanism. By "habit formation" we mean this functional, so "habit formation exists" is a claim about the sign of u_H, not a claim that habituation has been empirically distinguished from every alternative mechanism that can generate the same functional representation. This functional-level focus follows standard economic practice: preference functionals, rather than the psychological mechanisms that generate them, are the primitive objects of economic theory, and measuring them is the standard target of structural estimation and welfare analysis. Besides habituation, the same functional can arise from a range of other backward-looking mechanisms, including backward-looking reference dependence (Bowman et al., 1999), memory-based utility (Mullainathan, 2002), and consumption-dependent discounting (Shi and Epstein, 1993). For the choices and applications studied here, such mechanisms can be observationally equivalent and interchangeable; following Crawford (2010), we view distinguishing among them as beyond the paper's scope. We retain the label "habit formation" because the macroeconomic and financial literatures in which this functional plays a central role recognize it under that name. Alternatives outside this equivalence class include, notably, expectations-based reference dependence (Kőszegi and Rabin, 2006) and preferences over properties of consumption sequences (Loewenstein and Sicherman, 1991; Loewenstein and Prelec, 1993), as well as response heuristics unrelated to preferences. Appendix F.7

tion of H imposes neither case: under a general $u(C, H)$, the scale s is carried by $-u_H/u_C$, which equals s in the additive case. Its estimate in Section 3.3, 0.63 with a 95% confidence interval of [0.57, 0.71], lies below one, so habit is responsive to weaning.

[6] To see this, write a smooth evolution as $\dot{\mathcal{H}} = f(C, \mathcal{H})$ with $f = 0$ in every steady state. Differentiating along the diagonal gives $f_C = -f_{\mathcal{H}}$ at any steady state, so the first-order expansion is $\dot{\mathcal{H}} = \theta(C - \mathcal{H})$ with $\theta = -f_{\mathcal{H}}$. For linear f this expansion is exact, so θ is the exact decay rate for the linear specifications used in the literature; for nonlinear evolutions, such as that of Campbell and Cochrane (1999), θ is a steady-state decay rate.

examines these against the observed response patterns.

To provide micro evidence on habit formation, it is essential to determine whether and how habit influences utility. Accordingly, we measure six relevant preference parameters: the existence of internal habit formation, the existence of external habit formation, the habit decay rate (θ), the time discount rate (ρ), and two utility-derivative ratios ($-\frac{u_H}{u_C}$ and $\frac{u_{C_{\text{others}}}}{u_H}$).[7] The habit decay rate (θ) determines how rapidly habit adjusts to changes in consumption, while $-\frac{u_H}{u_C}$ captures the welfare impact of habit formation relative to consumption. $\frac{u_{C_{\text{others}}}}{u_H}$ quantifies the relative welfare impact of peer effects versus habit formation.

The measurement proceeds in three steps that the rest of the paper follows. First, for each point-identified parameter, a proposition converts a respondent's binary ranking of two spending profiles into an inequality bounding their parameter around a threshold determined by the spending amounts shown; Proposition 1, for instance, bounds the habit decay rate using the displayed spending deviations. The two existence parameters are sign-identified, since existence is a question of sign: the ranking directly reveals whether habit raises or lowers utility, so the only threshold is zero. Second, for the point-identified parameters, a follow-up question with a second threshold refines the inequality to one of four intervals with known thresholds, following the unfolding-bracket design of Barsky et al. (1997). Third, the statistical model of Section 2.5 treats each respondent's interval as an interval-censored observation of a latent parameter drawn from a population distribution, and estimates the population mean, which Appendix B shows is the corresponding parameter of the representative agent. The point estimates reported in the paper therefore come from the frequencies with which respondents fall into these intervals, not from any single bound.

2.2 Survey Experimental Design

Our survey is structured as follows: (i) consent form; (ii) demographic questions; (iii) calibration questions;[8] (iv) instructions on spending graphs with comprehension checks; (v) instructions on the hypothetical situation with comprehension checks; (vi) four modules on internal habit formation, habit decay rate, slope of indifference curve, and time discount rate; (vii) instructions on spending graphs under uncertainty; (viii) three modules on higher-order habit parameters;[9] (ix) instructions on spending graphs with other people's spending; (x) three modules on external habit formation, external habit mixture coefficient, and $\frac{u_{C_{\text{others}}}}{u_H}$; (xi) comprehension checks on the hypothetical situation (repeated); (xii) calibration questions (repeated); and (xiii) exit questions on respondents' approach to the survey and willingness to participate in a follow-up wave. Appendix H presents screenshots of the full survey.

[7] Estimating certain preference parameters requires knowledge of the time discount rate, ρ, whose elicitation is relegated to Appendix G.1 given its indirect relevance.

[8] Calibration questions are survey questions designed to have the same objective state across all respondents, enabling the identification and measurement of scale-use heterogeneity across individuals (Benjamin et al., 2023).

[9] Modules (vii) and (viii), which measure second-order utility derivatives, are designed to address distinct research questions (e.g., testing additive and multiplicative habits) and are deferred to a companion paper. Module (x) includes questions to measure an external habit mixture coefficient, which is omitted from this paper given the lack of evidence supporting external habit formation (Section 3.1).

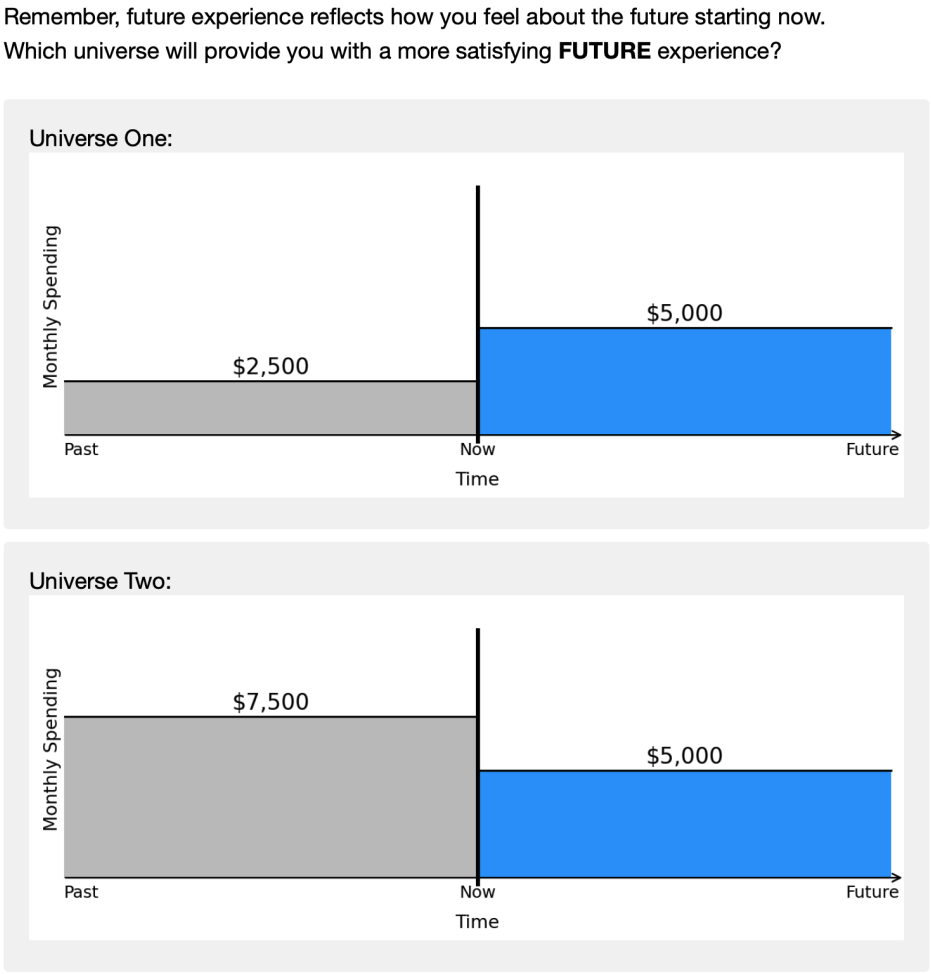


Figure 1: A Typical Survey Question

Past spending shapes habit, which in turn may influence well-being, suggesting that varied spending profiles can produce distinct welfare outcomes, with preference parameters modulating the process. Our core identification strategy thus involves recovering preference parameters from respondents' welfare rankings of spending profiles.[10] Correspondingly, our main survey questions ask respondents for their binary rankings of spending profiles designed to identify parameters of interest. The identifying variations are parameter-specific and are described in Section 3, where the contextual richness facilitates a clearer discussion of the specific identification strategies.

To enhance respondents' intuitive comprehension and facilitate comparisons between spending profiles, visual representations of these profiles were employed (see Figure 1 for two examples). In these graphs, the horizontal axis represents time, with the past on the left, now in the center, and the future on the right. Monthly spending is depicted as bars above the time axis, scaled proportionally to the spending amounts

[10]Because income affects utility through spending, the experiments do not specify the income process except to inform respondents that they can afford the spending profiles in the survey.

and color-coded to distinguish between different temporal periods. For instance, the first spending profile in Figure 1 illustrates monthly spending of $2,500 in the past up to now, followed by an increase to $5,000 per month in the future starting now. To ensure accurate interpretation, respondents were provided with detailed instructions and underwent comprehension assessments regarding these graphical representations.

The baseline level of household spending is $5,000 per month, close to average household spending reported by the Bureau of Labor Statistics and familiar to our respondents, whose average household spending was $4,362 and $4,265 in the two survey waves, respectively. Several robustness checks show that deviations between hypothetical and actual spending do not drive the estimates. First, half of the respondents with monthly spending below $4,000 were randomly assigned a $3,000 monthly baseline spending; their estimates do not differ significantly from those of the default-baseline group (Column 1 of Table A.5). Second, past spending, expected spending, and the $3,000-baseline indicator exert no statistically significant impact on the estimates of the preference parameters at the 5% level after adjusting for multiple hypothesis testing (Appendix Table A.4)—a result that remains robust across a variety of specifications. Third, restricting the sample to respondents whose actual spending lies close to the baseline spending gives similar results (Column 2 of Table A.5).

The spending profiles presented in our survey involve variations in both the level and timing of spending, designed to mimic common, real-world household spending fluctuations—approximately 50% of U.S. households reported changes in monthly spending from the prior year in 2024 (Larrimore et al., 2025), and checking account data for over 250,000 households show that the typical median-income U.S. household experiences average month-to-month spending fluctuations of 29% (Farrell and Greig, 2017). In our experiments, the monthly spending varies no faster than the annual frequency, with the average variation kept within 35% of the baseline spending level. To simplify the variations while preserving identification, spending deviates from the baseline only across two stylized time frame categories: (1) all past or all future years and (2) the last or next year. When questions involve other people's spending, we exclusively use the simpler variation category (1). The survey leaves horizons in category (1) deliberately open, telling respondents that the past means as far back in time as they can remember and the future as far ahead as they can imagine, with thirty years offered only as an optional anchor. Appendix F.2 shows that the thresholds identifying the existence of habit formation and the habit decay rate do not depend on the horizon, and that the remaining estimates are insensitive to it. The levels of spending deviations were chosen to represent meaningful economic changes and were refined through pilot studies.[11] The specific combinations of spending time frames and levels were guided by the identification propositions (e.g., Propositions 1–3), ensuring that the elicitation covers an informative part of the domain for each parameter of interest. Finally, comprehension checks (in module iv) confirm that respondents effectively engaged with the spending variations: 58% of respondents passed the checks on their initial attempt, and the remaining 42% passed on their second attempt.

[11] Pilot studies indicated that respondents found absolute-level framing of spending variations more intuitive and easier to evaluate than growth-rate framing; we therefore use absolute levels throughout.

Hypothetical scenarios involving future spending are routinely used in preference elicitation (e.g., Barsky et al., 1997; Ameriks et al., 2020). Since habit is shaped by the past, the identification of its parameters requires variations in past spending. Our experimental approach of constructing hypothetical scenarios involving past spending, while new in this context, is firmly grounded in counterfactual thinking—a fundamental cognitive process ubiquitous across human reasoning, such as learning, problem-solving, and causal inference (Epstude and Roese, 2008). This approach is exemplified by intuitive reflections such as, "If I had spent less back then, I would be in a better place now," illustrating how people readily contemplate historical variations. Respondents' open-ended explanations of their choices show that they can evaluate these counterfactual past variations (Section 3.1). Two clarifications delimit what this approach requires. First, the main object respondents rank is anticipated utility. Because the core questions ask which profile yields a more satisfying *future* experience, the evaluation is prospective—representing the assessment of future experience that governs forward-looking choice and that intertemporal-choice and asset-pricing models require. Second, identification does not require that respondents perform this ranking without error: random response errors are explicitly modeled and accounted for through the two-wave design (Section 2.5), and the estimates remain valid provided that residual biases do not systematically reverse the ranking (Section 2.3).

To help distinguish the spending profiles specified in the questions from respondents' own spending histories, the survey employs a parallel-universe analogy in which all factors are identical except for the spending profiles. Respondents demonstrated a strong grasp of this analogy, with 93% passing the relevant comprehension check on the first attempt. Respondents then choose which universe would yield a more satisfying past or future experience—how they feel about the past up to now or the future starting now. Figure 1 illustrates a typical question.

To reduce cognitive load and control for potential confounding factors, while preserving identification, respondents' choices are situated within the following hypothetical scenario (with emphasis as in the survey):

To simplify the comparison of your experience under various spending patterns, please evaluate your experience in the following hypothetical situation:

- *There is* ***no inflation****, and prices of everything stay the same over time.*
- *You* ***rent the durable goods*** *you use, including residence, furniture, car, etc.*
- *Your* ***preferences do not change*** *over time.*
- ***People not mentioned in the questions always spend $5,000 per month****, regardless of their income or wealth.*
- ***Any other factors not specified in the questions are identical between the universes and remain identical over time.***

The hypothetical situation alleviates respondents' concerns regarding changes in the purchasing power of money, the acquisition of durable goods, and potential shifts in preferences. It is standard practice to exclude inflationary dynamics from hypothetical spending scenarios when inflation is not the primary focus of in-

vestigation (e.g., Kapteyn and Teppa, 2003). The acquisition of durable goods can be seen as antithetical to habit formation, as the former tends to increase utility, whereas the latter reduces it. Renting durable goods mitigates this confounding effect. Consistent with the scenario, the literature finds that people typically operate under the assumption of preference stability (e.g., Loewenstein et al., 2003). Because the spending of others may influence individual well-being, the hypothetical situation controls for this factor when it is not part of the identifying variation. The scenario holds all other factors constant across universes and over time. To reinforce this point, the instructions clarify that "*The last point means that factors, including but not limited to your income, savings, wealth, control over finances, and the state of the economy, are the same between the universes and remain the same over time.* ***Only your spending mentioned in the questions varies between the universes****, so please focus solely on those spending differences when making your evaluations.*"[12] This ceteris-paribus instruction is a structural feature of preference elicitation that traces back at least to Barsky et al. (1997), whose foundational measurements of preference parameters based on hypothetical-choice questions likewise required respondents to hold non-treatment factors implicitly constant; we assess its effectiveness in our setting through the comprehension checks and open-ended explanations documented below. The instructions explicitly define "you" as your household and "others" as other households. Both entities are described as being able to afford the spending specified in the survey questions.

The abstraction in our hypothetical scenarios is a deliberate choice to minimize hypothetical bias and sharpen identification. Adding realistic but irrelevant details, such as inflation dynamics, would raise cognitive burden and reduce engagement, which tends to amplify, rather than reduce, hypothetical bias (Haghani et al., 2021b). Leaving confounds uncontrolled risks mixing habit formation with factors such as ownership of durable goods. Our design therefore varies total consumption while holding other factors fixed, matching how people naturally evaluate counterfactuals by changing only the key variable of interest.

To ensure comprehension, key aspects of the hypothetical situation are bolded in the survey (as shown above), and respondents had to pass comprehension tests on both the hypothetical scenario and the monthly spending graphs before reaching modules on preference parameters. For the comprehension checks on the hypothetical situation (module v), 93% of the participants passed on the first attempt and the remaining 7% passed on the second attempt. To assess sustained understanding of the situation, the survey repeated the same checks near the end (in module xi): 92% answered all six questions correctly, 5% missed one, and 3% missed two or more.

The survey was described to respondents as a study of spending behavior, which is an accurate non-specific characterization of the binary-choice task, rather than as a study of habit formation specifically.

[12] To address potential confusion regarding the relationship between income, saving, and spending, the survey instructions include a clarifying note: "*Even though your income, saving, and wealth are the same across universes, your spending doesn't have to be identical. For the purposes of this survey, you can imagine a hypothetical external financial system that automatically adjusts your savings at no cost to you, allowing your spending to vary while keeping everything else unchanged.*" This is a deliberate design choice to isolate effects from consumption, free from confounding wealth effects or savings optimization. While this might be confusing for some respondents, we can detect such misunderstandings using open-ended explanations of respondents' choices, and our results are robust to removing confused respondents (Appendix F.5).

This framing was chosen for three reasons. First, to mitigate priming bias: revealing the focus on habit formation could undermine the validity of the test for its existence. Second, to prevent confusion, since the average respondent is unfamiliar with habit formation as modeled in economics. Third, to minimize selection on the survey topic. This practice is standard in preference measurement studies, which are routinely described to participants by their broad domain rather than by the specific construct under measurement. This non-specific framing also limits experimenter demand: when cues about expected answers are weak or implicit, demand effects are likely small (de Quidt et al., 2018).

The main survey questions use binary choices to reduce cognitive load. To enhance the precision of the estimates for point-identified parameters, an adaptive unfolding bracket methodology is implemented.[13] This approach entails an initial ranking of two spending profiles, followed by a second ranking in which one or both profiles are adjusted based on the respondent's first choice, narrowing the range of parameter values consistent with the responses. An illustration of this technique is provided in Section 3.2. The binary-choice format deliberately excludes an option for indifference, thereby reducing the propensity for satisficing behavior (Caplin et al., 2011) and fostering more thorough engagement with the presented scenarios.[14] When respondents perceive the options as equivalent, their selections are presumed to be stochastic, consistent with stochastic choice theory (Block and Marschak, 1960).

The main survey questions elicit preferences rather than objectively correct or incorrect answers, so direct incentivization is infeasible. To counteract potential biases arising from misinterpretation, inattention, or protest responses, the survey instrument was designed to minimize cognitive load: visual spending profiles, repeated emphasis that spending is the only difference between response options, explicit statement of when the two spending profiles differ, and follow-up questions with dynamic explanations of spending changes. To facilitate the distinction between past and future experiences, survey questions solicited responses for both, with definitions reiterated and the words "past" and "future" highlighted in each question. Interactive answer options darken when hovered over or selected, helping respondents avoid accidental clicks.

To ensure sustained participant attention throughout the survey, a layered approach incorporating both explicit and implicit attention checks was implemented. Explicit checks consist of initial comprehension questions and targeted questions with specific expected responses. Implicit checks include response time, exit comprehension checks, and cross-wave response consistency. Participants knew that attention checks existed but not where they appeared. To further incentivize diligent participation, high-quality responses entered into a lottery with a 1 in 100 probability of winning $5.[15] The survey also asked respondents to self-report attention and effort: over 99% of participants reported maintaining full attention midway through

[13] This design diverged from Ameriks et al.'s (2020) strategic survey question format because the identifying variations encompass both spending levels and time horizons. This multifaceted structure raises the cognitive burden, rendering their approach impractical for eliciting indifference points here.

[14] The sole exception pertains to the time discount rate question regarding past experiences, where the spending profiles across options were objectively identical, and participants were offered the opportunity to indicate indifference.

[15] From the 2,097 valid responses, 21 respondents were randomly selected for this award.

the survey. At the end, 82% reported maximum effort, 16% substantial effort, and 1% minimal effort.

To further enhance reliability, we follow Barsky et al. (1997) and conduct the survey in two waves, with the first wave fielded in August 2024 and the second wave two weeks later in September 2024. In the second wave, the core modules were reordered, and the order of response options was reversed to mitigate order-related response biases. To reduce respondent burden, the second wave omitted the demographics module and the two modules of calibration questions. The baseline spending level from the first wave was retained in the second wave. The test-retest correlations for the core module responses averaged 0.45 (see Appendix Table A.1 for parameter-specific correlations across the waves),[16] which falls within the typical reliability range (0.15–0.60) for preference measurements through choice experiments (Chuang and Schechter, 2015). Estimates using only first-wave data (Column 5 of Appendix Table A.5) are quantitatively close to the benchmark estimates, indicating that the second wave mitigates response error and enables the test-retest validity check, rather than driving the estimates.

It is worth emphasizing that comparisons of the spending profiles do not amplify the significance of habit formation. Comparative evaluations do not intrinsically result in estimates indicative of habit formation ($u_H < 0$); these evaluations are equally compatible with $u_H \geq 0$, under which habit formation is not present. Importantly, neither the experimental design nor the estimation procedure constrains the sign of u_H, thereby allowing the data to reveal the genuine effect of habit formation without introducing bias. Whereas real-world behavior is frequently subject to many intersecting factors that obscure the influence of habit formation, the controlled experimental environment is deliberately structured to isolate consumption patterns needed to identify key habit parameters, thus providing evidence that is difficult to obtain from real-world contexts.

2.3 Validity of Hypothetical Choices

Systematic deviations between hypothetical and real-world choices are typically categorized as hypothetical bias (Haghani et al., 2021a). Our design therefore targets the most likely sources of hypothetical bias in the context of eliciting preferences over total-consumption profiles.

The main threats in this setting are cognitive bias and lack of consequentiality (Haghani et al., 2021a; Johnston et al., 2017). Cognitive bias is the sharper concern: respondents in an unemotional "cold" state may mispredict utility under different consumption profiles. To mitigate this concern, our experimental design anchors the scenarios to empirical realities, a strategy known to enhance the validity of elicited responses (Manski, 2004; Haghani et al., 2021b). Specifically, the baseline monthly household spending level of $5,000 closely approximates average U.S. household expenditures reported by the Bureau of Labor Statistics, with 85.9% of respondents' household spending lying within one standard deviation

[16] The test-retest correlation for the responses on external habit formation was excluded from this calculation as the absence of evidence on external habit formation (Section 3.1) suggests that their responses are random and thus naturally have low cross-wave correlation.

from this baseline. For respondents with lower actual spending, we randomized half of them into a $3,000 baseline spending and find that the estimates from this group are statistically indistinguishable from those of the main sample (Appendix F.1), indicating that the default baseline spending level does not bias responses. The variations in our consumption profiles also mirror the common life experiences of U.S. residents (Section 2.2). Because consumption choices are familiar, respondents can draw on well-formed preferences, which reduces hypothetical bias (List, 2001). Their open-ended responses also show effective counterfactual reasoning about past spending variations (Section 3.1).

Crucially, our approach does not require respondents to report absolute utility levels. Our method remains valid provided that cognitive biases do not alter the ranking of options—an assumption that holds if biases affect all predictions symmetrically. This distinction is well supported by the literature, which suggests that hypothetical bias tends to distort absolute levels while preserving the reliability of relative trade-offs (see Haghani et al., 2021a, for a review).[17] This ranking assumption carries testable implications that the data support: respondents' open-ended explanations of the existence question reflect the intended trade-offs (Section 3.1), and the elicited parameters exhibit convergent and predictive validity (discussed below and in Sections 3.2–3.3)—patterns that would be difficult to generate if biases systematically reversed the elicited rankings. Since choices are not financially binding, respondents may exert insufficient effort. We address this through several mechanisms detailed below, including comprehension checks and incentivizing response quality.

Other common sources of hypothetical bias, such as social desirability or strategic behaviors, are less pertinent here. Social-desirability bias is most concerning for sensitive self-reports—criminal, health, moral, voting, and some economic outcomes (Bursztyn et al., forthcoming). Our task asks respondents to rank total-consumption profiles rather than disclose stigmatized behavior or socially judged traits, so the scope for image-driven misreporting is limited. The academic nature of our survey, which was not framed as a commercial or policy pricing study, makes it unlikely that respondents would strategically misrepresent their preferences to influence a market or policy outcome.

To counter these potential biases, we built a suite of ex-ante mitigation strategies into the survey instrument, focusing on contextual realism, cognitive load reduction, and mitigation of inattention and indifference. By incorporating clear visual representations of spending profiles, a "parallel-universe" analogy, and baselines tied to average consumer expenditures, we mitigate the "hot-cold empathy gap" and

[17] Projection bias (Loewenstein et al., 2003)—the tendency to project one's current circumstances onto the evaluation of counterfactual conditions—is one example of such a bias: consistent with the distinction drawn above, it distorts the perceived *level* of utility rather than the *ranking* of the universes, so even though it is directional it does not by itself reverse the sign of u_H. Our existence results are therefore robust to it, and at most the estimated *magnitude* of habit decay could be affected. That direction is itself unsettled in our setting, in which respondents mainly evaluate future experience under counterfactual past spending: while a projection or anchoring channel would make our estimate a lower bound on the true decay rate (widening the discrepancy with the asset-pricing–calibrated dynamics in Appendix E), memory-based channels such as fading-affect bias (Walker et al., 1997)—operating through the counterfactual past—would push toward an upper bound, so we do not rest the magnitude on a presumed direction. Reassuringly, projection bias operates through the gap between a respondent's actual circumstances and the hypothetical scenario; the $3,000-baseline arm narrows that gap for lower-spending respondents—leaving less room for current-state projection—yet yields estimates statistically indistinguishable from the main sample.

limited familiarity, creating a tangible decision-making environment to strengthen internal and external validity. To minimize reliance on simplifying heuristics, we employed a streamlined binary-choice format, provided detailed instructions with comprehension checks, and controlled for complicating factors such as inflation, thereby reducing cognitive burden and encouraging deliberate responses. Additionally, explicit and implicit attention checks, a response-quality lottery, and no indifference option discourage inattention and satisficing. The two-wave design separates preferences from transitory response error. Appendix F shows the robustness of our estimates across demographics, spending baselines, time horizons, response biases and errors with nonzero and wave-varying means, first-wave-only estimation, strict compliance with the ceteris-paribus structure of the hypothetical situation, and additional attention checks.

Given the lack of readily observable counterparts for these structural habit parameters in conventional data, we assess the validity of our hypothetical choices by demonstrating their convergent and predictive validity (Strauss and Smith, 2009). To establish convergent validity, we show consistency of our estimates with findings on related behaviors in the literature: the estimated rate of habit decay closely aligns with prior estimates of income adaptation speed (Section 3.2), and the derived welfare impacts of habit formation and peer effects effectively match established estimates for income adaptation and peer effects, respectively (Section 3.3). These benchmarks are not competing estimates of the parameters measured here. What the adjacent literatures report are downstream implications of those parameters in other domains, namely how quickly reported well-being adapts to income and how much others' spending reduces that well-being, recovered from different data by different methods. Agreement therefore tests the instrument rather than duplicating the measurement, and licenses its use for the parameters with no direct estimate to compare against, notably the decay rate of the total-consumption habit and the relative welfare impact of habit formation and peer effects. Qualitative responses provide further support: about 80% of respondents whose choices indicate internal habit formation cited reasons reflecting the difficulty of cutting back from accustomed spending (Section 3.1). To demonstrate predictive validity, we show that the habit decay rate measured in our survey predicts real-world happiness responses: respondents who reported faster habit adjustment in our hypothetical scenarios also exhibited faster hedonic adaptation to the 2024 U.S. presidential election outcome, which occurred two months after our survey (Section 3.2).

2.4 Data Summary

The survey was administered via Prolific, an online platform designed for human intelligence tasks, using its representative-sample feature. To mitigate cultural bias, the sample was restricted to residents of the United States. While online samples may not achieve the representativeness of census-based samples, they offer greater representativeness than in-person convenience samples (Berinsky et al., 2012). These samples are commonly employed in economic research (e.g., Kuziemko et al., 2015; Bordalo et al., 2016; Kumar et al., 2023; Sergeyev et al., 2024). Beyond this, research indicates that Prolific yields higher response

quality relative to other online platforms (Peer et al., 2022; Douglas et al., 2023). Median participants spent about 31 and 23 minutes on the two-wave surveys, earning $4.70 and $2.70, respectively, for an effective hourly rate of approximately $8.00.

Of the 1,367 completed first-wave responses, 1,243 respondents indicated willingness to participate in a follow-up, and 730 (approximately 59%) completed the second wave. Completion of the survey required passing the in-survey comprehension checks on the spending graphs and hypothetical scenario, as well as at least one of two instruction manipulation checks—used here as attention checks. Responses that did not meet these requirements, or were abandoned mid-survey, were not in the sample by construction. The benchmark estimation includes completed responses from both waves; Appendix F.6 shows that the estimates are robust to dropping respondents who failed one of the two manipulation checks.

Attrition between waves raises the question of whether the respondents who returned differ from those who did not. Three considerations address it. First, on observables the returners closely resemble the full first-wave sample: no category in Appendix Table A.3 differs by as much as four percentage points between the first- and second-wave columns. Second, the likelihood in Section 2.5 uses every first-wave response and sets the contribution of a missing second-wave response to one, so each population mean is anchored by the full first wave, with the second wave adding precision and identifying response error. Third, and most directly, estimates using only the first wave are free of attrition by construction and quantitatively close to the benchmark (Column 5 of Appendix Table A.5), indicating that any selection into the second wave does not materially move the reported parameters.

Appendix Table A.3 presents the sample's demographic distributions by wave alongside those of the 2020 U.S. Census. The sample is broadly representative on gender, race, household size, and region, but, as is common in online samples, older individuals (65+), those with less than some college education, and those with incomes exceeding $120,000 are represented differently than in the Census. Appendix F.1 addresses these differences by allowing the estimated preference parameters to vary with observed demographics; none of the demographic coefficients are statistically significant after adjusting for multiple hypothesis testing.

2.5 Statistical Model

Following Barsky et al. (1997), this study employs a statistical model to extract consistent variations across the two waves of responses and to mitigate response errors not addressed by the survey experimental design. The model follows naturally from the structure of the responses. Each response places a respondent's parameter inside a bracket whose endpoints are known but unequally spaced, so the parameter must be estimated from both the endpoints and the share of respondents falling in each bracket, rather than by averaging response categories. The two waves then discipline the error structure: their cross-wave covariance identifies the transitory component of response error and supports the test-retest and wave-varying-mean checks.

Potential remaining response biases and errors are addressed through robustness checks in Appendix F.

The observed response for preference parameter x can vary across respondents, indexed by i, and waves, indexed by w, and is generated by $X_{i,w} \equiv \sum_{k=1}^{K} k \cdot 1\left(T_{k,\tilde{x}} < \tilde{x}_{i,w} \leq T_{k+1,\tilde{x}}\right)$, where k indexes response options and $1\left(\cdot\right)$ is the indicator function. The latent variable is $\tilde{x}_{i,w} = x_i + \varepsilon_{i,x,w}$. $T_{\{k\},\tilde{x}}$ denotes the sequence of known thresholds determined by the identifying propositions (Propositions 1–4) and the experimental variations (see Table 4 for a summary and Section 3 for details), with $T_{1,\tilde{x}} = -\infty$ and $T_{K+1,\tilde{x}} = +\infty$. The true parameter value for individual i, x_i, is drawn from $\mathcal{N}(\mu_x, \sigma_x^2)$. The response error, $\varepsilon_{i,x,w}$, can vary across respondents, parameters, and waves, and is drawn from $\mathcal{N}(0, \sigma_{\varepsilon_x}^2)$ independently of the true parameter value for each wave. A robustness check allows the means of the response errors to be nonzero and to vary across waves (Appendix F.3), finding that the estimated means are indistinguishable from zero and that the estimates of the preference parameters are not significantly different from those under the specification here. For aggregation and computation, the parameters are assumed to be independent within a respondent. Because the respondents are geographically dispersed across the United States and are unlikely to be acquainted, the responses are assumed to be independent across respondents. Allowing the response errors to persist across waves (i.e., $\sigma_{\varepsilon_{x,P}}^2 = Cov\left(\varepsilon_{i,x,1}, \varepsilon_{i,x,2}\right)$ and $\sigma_{\varepsilon_x}^2 = \sigma_{\varepsilon_{x,P}}^2 + \sigma_{\varepsilon_{x,T}}^2$, where P and T indicate the persistent and transitory components, respectively), we arrive at the joint distribution of respondent i's parameter $\tilde{x}$ in the two waves of the survey:

$$\begin{bmatrix} \tilde{x}_{i,1} \\ \tilde{x}_{i,2} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_x \\ \mu_x \end{bmatrix}, \begin{bmatrix} \sigma_x^2 + \sigma_{\varepsilon_{x,P}}^2 + \sigma_{\varepsilon_{x,T}}^2 & \sigma_x^2 + \sigma_{\varepsilon_{x,P}}^2 \\ \sigma_x^2 + \sigma_{\varepsilon_{x,P}}^2 & \sigma_x^2 + \sigma_{\varepsilon_{x,P}}^2 + \sigma_{\varepsilon_{x,T}}^2 \end{bmatrix} \right),$$

whose cumulative distribution function is denoted by Φ.

Given the predominance of representative-agent frameworks in contemporary habit formation models, this study examines the implications of parameter estimates within such models. In Appendix B, we establish that individual agents' parameter values aggregate to their population mean for the representative agent. Formally, let x_R denote the representative agent's value for parameter x, and let x_i represent the corresponding parameter value for individual i. We show that $x_R = \frac{1}{N}\sum_{i=1}^{N} x_i$, where N is the number of individuals. Since $\frac{1}{N}\sum_{i=1}^{N} x_i \xrightarrow{p} \mu_x$, the parameter of interest in estimation is μ_x, the mean of the individual parameter values.[18]

Estimates are derived from joint maximum likelihood estimation, with the likelihood function

$$\prod_{i,x} \left[\Phi\left(\begin{bmatrix} T_{X_{i,1}+1,\tilde{x}} \\ T_{X_{i,2}+1,\tilde{x}} \end{bmatrix} \right) - \Phi\left(\begin{bmatrix} T_{X_{i,1}+1,\tilde{x}} \\ T_{X_{i,2},\tilde{x}} \end{bmatrix} \right) - \Phi\left(\begin{bmatrix} T_{X_{i,1},\tilde{x}} \\ T_{X_{i,2}+1,\tilde{x}} \end{bmatrix} \right) + \Phi\left(\begin{bmatrix} T_{X_{i,1},\tilde{x}} \\ T_{X_{i,2},\tilde{x}} \end{bmatrix} \right) \right].$$

For respondents observed only in the first wave, the second-wave response is treated as missing and its probability contribution is set to one, so their likelihood contribution reduces to the corresponding

[18] σ_x^2 is not separately identified, since the cross-wave covariance pins down only $\sigma_x^2 + \sigma_{\varepsilon_{x,P}}^2$; this does not affect identification of the parameter of interest μ_x.

first-wave marginal probability. To address the computational challenges posed by joint estimation, we adopt a Bayesian approach, thereby circumventing the need for direct maximization of the likelihood function. Specifically, we employ Hamiltonian Monte Carlo, a Markov Chain Monte Carlo method known for its superior sampling efficiency, together with uniform priors for the main parameters of interest to ensure equivalence between posterior modes and maximum likelihood estimates.

3 Measurement

This section details the measurement of the preference parameters of interest, including the identifying variations, survey questions, and resulting estimates.

3.1 Existence of Habit Formation

Existence of internal habit formation. Under the functional $u(C, H)$, habit formation is the property that a higher habitual level of spending lowers current utility; that is, $u_H < 0$. Because habit, H, increases with past consumption, internal habit formation is consistent with the utility differential $Q_{HI} \equiv u(C, H + \Delta h) - u(C, H)$ being negative for $\Delta h > 0$ stemming from higher past own consumption, but is inconsistent with a non-negative differential.

To identify the sign of Q_{HI}, it is thus intuitive to vary the respondent's past spending while controlling for future spending, so that variation in future experience comes only from variation in habit. Figure 1 shows a screenshot of the resulting survey question eliciting the sign of Q_{HI}.[19] In this question, preferring a spending profile with low past spending (low H) over one with high past spending (high H) for a better future experience implies $Q_{HI} < 0$. In other words, choosing Universe One in the question shown in Figure 1 is consistent with internal habit formation, while choosing Universe Two is not. It is worth emphasizing again that the survey does not prime respondents with habit formation and that no assumption is made about the sign of Q_{HI}. Additionally, comparison (over time) does not necessarily imply $Q_{HI} < 0$, because the comparison could also be consistent with $Q_{HI} \geq 0$, in which case internal habit formation would not exist.

Response distributions to this question show that about two-thirds of respondents chose lower past spending for a better future experience—Universe One—in each wave of the survey (Panel A of Table 1), consistent with the existence of internal habit formation. After accounting for response errors, the estimate of $\text{sgn}(Q_{HI})$ corroborates this finding (Table 2): it is -1, and its 95% confidence set, which collects the sign values not rejected at the 5% level, contains only -1. The two-thirds majority is also difficult to reconcile with expectations-based reference dependence (Kőszegi and Rabin, 2006), which predicts an even split in this question because future spending is identical across the two universes and fully anticipated

[19] In Figure 1, past spending deviates by \$2,500 below or above the baseline spending of \$5,000. In the survey version with the baseline spending of \$3,000, past spending varies by \$1,000 below or above the baseline.

Table 1: Response Distributions

	First wave				Second wave			
Panel A: Sign-identified parameters	U1		U2		U1		U2	
Existence of internal habit formation	64.3%		35.7%		65.3%		34.7%	
Existence of external habit formation	51.1		48.9		53.2		46.8	
Panel B: Point-identified parameters	U1U1	U1U2	U2U1	U2U2	U1U1	U1U2	U2U1	U2U2
Habit decay rate	34.7%	19.5%	18.5%	27.4%	39.0%	18.5%	16.0%	26.4%
$-u_H/u_C$	34.1	8.6	16.9	40.5	33.4	7.3	18.6	40.7
$u_{C_{\text{others}}}/u_H$	48.1	14.0	6.8	31.0	47.5	14.5	10.0	27.9

Notes: Sample: 2,097 observations. Entries are percentages. UX: Universe X. U1U2: chose Universe One in the initial question and Universe Two in the follow-up; analogous notation applies to other response sequences. The second wave reversed the order of the two universes (its Universe One is the first wave's Universe Two); for comparability, responses from both waves are reported with first-wave labels.

in each; Appendix F.7 examines this and other alternative interpretations.

At the conclusion of the survey, respondents provided open-ended explanations of how they arrived at their answers to the question regarding the existence of internal habit formation.[20] The full question and their choices were displayed. Using three LLM classifiers with human adjudication of disagreements (Appendix D), we categorize the explanations and rank them by frequency, following a growing literature that uses open-ended survey responses to uncover the mechanisms behind choices (Haaland et al., 2025). Table 3 presents the most common explanations.

Since the precise concept of habit formation typically lies outside conscious awareness (Wood and Rünger, 2016), respondents might not accurately attribute their choices to it. Therefore, we interpret responses such as "increased (future) spending (relative to past) leads to (more) satisfaction" as evidence that habit formation influenced their decisions. For instance, one respondent noted, "It's the same amount of money, but because I spent way more in the past in Universe Two than One, the amount in One would FEEL like more than it's in Two. Spending more is never a problem; it's having to cut down from spending that's hard" (respondent's emphasis). Another stated, "Because I'm accustomed to spending less ($2,000) a month, if I have more money to spend, my future will be happier and more satisfying." As shown in Table 3, about 80% of respondents who selected Universe One ($Q_{HI} < 0$) cited reasons consistent with habit formation, while the remaining 20% indicated misunderstanding of the hypothetical scenario (e.g., incorrectly assuming higher income or savings in Universe One) or general errors.

The reasoning of respondents who selected the option inconsistent with habit formation (Universe Two, $Q_{HI} > 0$) also provides insight. Approximately two-thirds of these respondents cited a preference for "decreased (future) spending (relative to past)," often associating it with frugality, better financial

[20]To maintain a reasonable survey duration, respondents were not asked to explain their responses to other questions. For those questions, we assess validity through the shared comprehension checks, along with the convergent and predictive validity checks where possible.

Table 2: Parameter Estimates

	Estimate	95% CI
sgn (Q_{HI})	-1	{-1}
sgn (Q_{HE})	1	{-1, 0, 1}
Habit decay rate	1.03	[0.87, 1.18]
Habit decay factor (annual)	0.64	[0.58, 0.69]
$-u_H/u_C$	0.63	[0.57, 0.71]
$u_{C_{\text{others}}}/u_H$	0.30	[0.13, 0.47]
$u_{C_{\text{others}}}/u_C$	-0.19	[-0.30, -0.08]
$u_H/u_C + u_{C_{\text{others}}}/u_C$	-0.82	[-0.96, -0.69]

Notes: Annual habit decay factor: $1 - e^{-\text{Habit decay rate}}$. $u_{C_{\text{others}}}/u_C$ and $u_H/u_C + u_{C_{\text{others}}}/u_C$ are implied by $-u_H/u_C$ and $u_{C_{\text{others}}}/u_H$. sgn $(\cdot)$ takes values in $\{-1, 0, +1\}$, and its 95% CI denotes the set of sign values not rejected at the 5% level.

control, or greater efficiency. While these reasons suggest an implicit misunderstanding of the hypothetical scenario—in which unspecified factors such as frugality, financial control, and efficiency remain constant across time and response options—they do not contradict habit formation. Other respondents chose Universe Two due to more explicit misunderstandings, such as ignoring the survey instruction to rent durable goods, misinterpreting the constancy of income and savings, or making general errors or random choices. These responses highlight a practical trade-off in survey design: more complete ceteris-paribus instructions can reduce misunderstandings but make the scenario longer and harder to process. They also suggest that responses inconsistent with internal habit formation largely reflect added considerations or misunderstandings, rather than reasoning that contradicts internal habit formation itself.

Chetty and Szeidl (2016) propose that consumption commitments, arising from the costly adjustment of illiquid durable goods such as residences and vehicles, provide an alternative explanation for consumption patterns typically ascribed to habit formation. To distinguish between these concepts, this study employs a hypothetical scenario in which respondents rent durable goods to minimize adjustment costs. It includes only responses from participants who demonstrate a clear understanding of the scenario in the comprehension checks, with approximately 94% of them also indicating such understanding in the open-ended feedback discussed above. The survey question uses symmetric deviations between past and future spending, thereby excluding residual psychological or non-psychological factors, captured by symmetric (e.g., quadratic) adjustment costs, as drivers of the observed responses. The open-ended feedback also indicates that the vast majority of respondents' choices are not driven by costs related to adjusting consumption commitments. These findings suggest that habit formation can influence behavior independently of consumption commitments in scenarios that minimize adjustment costs.

Existence of external habit formation. External habit formation is consistent with lower future utility resulting from higher past spending by others. Allowing H to increase with others' past consumption,

Table 3: Rationale and Proportion of Respondents' Choices

	Proportion		
Rationale	First wave	Second wave	Overall
Choose U1 ($Q_{HI} < 0$) because			
Increased spending leads to satisfaction	76.0%	81.5%	50.3%
More saving	8.0	5.0	4.5
More income in the future	5.5	7.8	4.1
Other reasons	10.6	5.7	5.7
Total	100.0	100.0	64.6
Choose U2 ($Q_{HI} > 0$) because			
Reduced spending leads to satisfaction	63.9	71.5	23.5
Durable spending	18.5	13.4	5.9
Increased saving in the future	10.9	6.7	3.3
Other reasons	6.8	8.3	2.6
Total	100.0	100.0	35.3

Notes: Primary rationales for choices in the internal-habit existence question. Entries are percentages. As in Table 1, responses from both waves are reported with first-wave universe labels.

external habit formation predicts that the utility differential $Q_{HE} \equiv u(C, H + \Delta h) - u(C, H)$ is negative for $\Delta h > 0$ arising from higher past spending by others, but is inconsistent with a non-negative differential.

To identify the sign of Q_{HE}, it is natural to vary other people's past spending while holding constant both their future spending and the individual's own past and future spending. This ensures that variation in future experience is driven solely by changes in habit due to other people's past spending. To visually represent these variations, the spending graphs were augmented with others' spending profiles. To facilitate comparisons over time and across people, one's own spending graph is compressed horizontally so that both one's own and others' spending graphs fit on a single line (see Figure 2). Different colors distinguish between the spending of oneself and others. Respondents received a brief instruction about these visual changes and were informed that the compressed graphs encompassed the same time frames as the uncompressed graphs.

In this context, preferring a spending profile with low past spending by others (low H) over one with high past spending by others (high H) implies $Q_{HE} < 0$. In other words, choosing Universe Two in the question is consistent with external habit formation, while choosing Universe One is not. Since the survey does not prime respondents with habit formation and no assumptions are made about the signs of derivatives of the utility function with respect to habit, comparison (over time and across people) does not necessarily imply $Q_{HE} < 0$ (that external habit formation exists), because the comparison could also be consistent with $Q_{HE} \geq 0$, in which case external habit formation does not exist.

The distribution of responses (Panel A of Table 1) suggests that respondents essentially choose randomly between the two universes in the question shown in Figure 2, providing no evidence that others' consumption systematically influences one's habit. After accounting for response errors, the estimate of sgn (Q_{HE}) includes 0 in its 95% confidence set (Table 2), indicating no evidence for external habit formation.

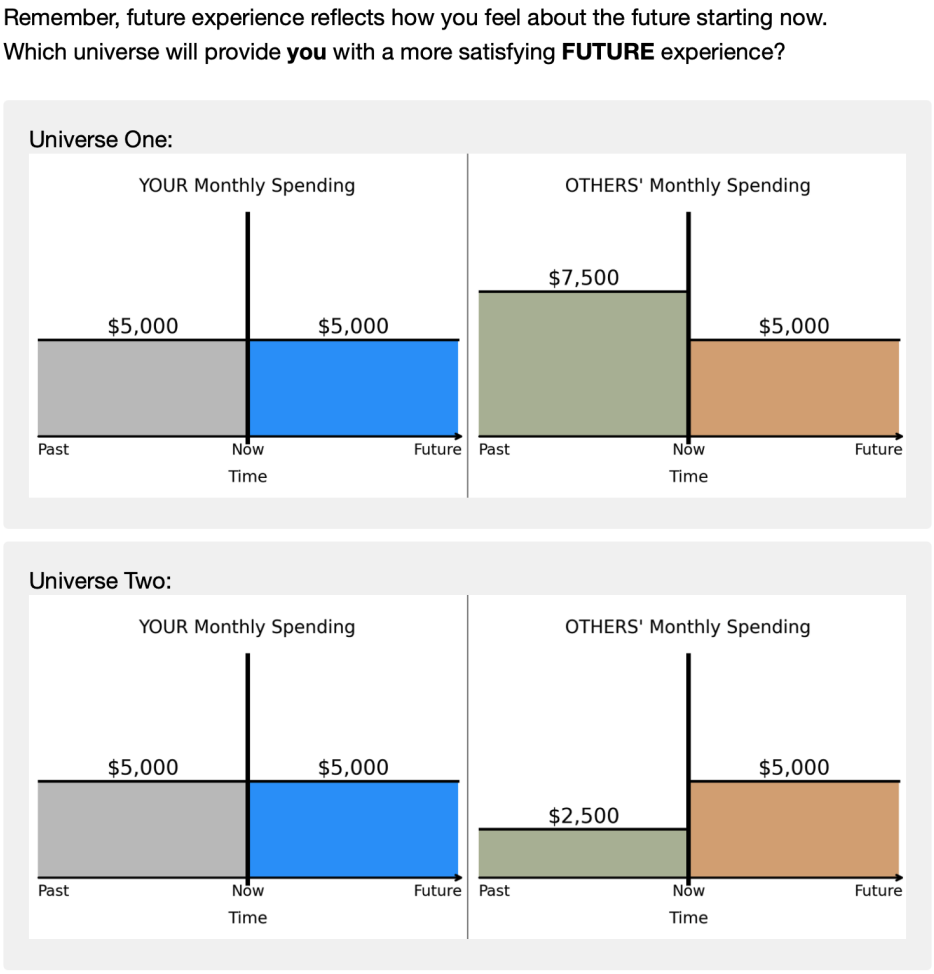


Figure 2: Survey Question on Existence of External Habit Formation

As discussed in Section 2.2, the survey questions exclude an indifference response option. The essentially uniform distribution of responses to this question supports this design choice, confirming that indifferent respondents tend to choose randomly in the absence of such an option. The finding of no evidence for external habit formation aligns with the evidence from Brunnermeier and Nagel (2008) and, combined with our evidence on internal habit formation, is consistent with models featuring internal habit formation while challenging those relying solely on external habit formation.

3.2 Speed of Habit Decay

The rate of habit decay is governed by θ, as in $\dot{H} = \theta\,(C - H)$. Since θ governs habit's persistence, a natural survey question to elicit it varies the persistence of past spending. To point-identify θ, the level of past spending must also vary to ensure a surjective mapping from θ to survey responses. Inverting this mapping recovers values of θ consistent with the responses, as formalized in Proposition 1. Appendix A provides proofs for all propositions.

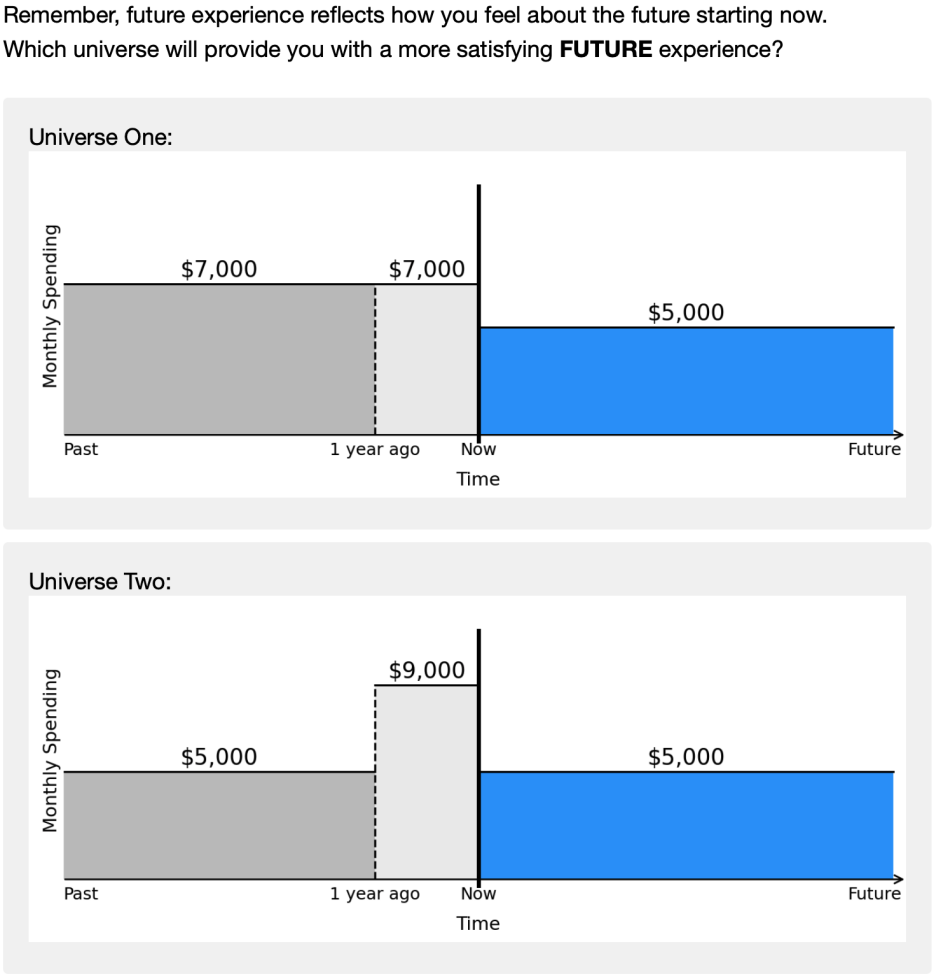


Figure 3: Survey Question on Speed of Habit Decay

Proposition 1. $\theta > -\ln\left(1 - \frac{\Delta C_{U1}}{\Delta C_{U2}}\right)$ *if a respondent chooses Universe One over Universe Two for a more satisfying future experience in a habit decay rate question.*[21]

Figure 3 presents a screenshot of the survey question for identifying θ. The question asks respondents to choose the spending profile that would yield a more satisfying future experience: either a medium-high level of spending spread over an extended period in the past (Universe One), or a very high level of spending concentrated in the recent past (Universe Two), while future spending remains constant. The intuition behind this identification strategy is that a preference for the profile with more persistent past spending (Universe One) implies that recent past spending contributes more to habit than distant past spending. This suggests that habit decays quickly, relative to the threshold $-\ln\left(1 - \frac{\Delta C_{U1}}{\Delta C_{U2}}\right)$, as specified in the proposition.

ΔC_{U1} and ΔC_{U2} denote the deviations in the levels of monthly spending from the baseline spending

[21]The paper's identifying propositions are stated as conditional statements, sufficient for its results, though all can be strengthened to biconditional (if and only if) statements. The notation of ΔC_{U1} and ΔC_{U2} is defined below.

level in Universe One and Universe Two, respectively. In Figure 3, $\Delta C_{U1} = \$2,000$ and $\Delta C_{U2} = \$4,000$. Thus, the survey question in Figure 3 partitions the possible values of θ into two complementary intervals, separated by the threshold $\ln 2$, which corresponds to an annual habit decay speed of 50%. Respondents selecting Universe One indicate a habit decay speed exceeding 50% per year, while those choosing Universe Two reveal a decay speed below this threshold.

Our design shares its identifying logic with the choice experiment that Rozen (2010) proposes for measuring habit decay, in which a difference in today's consumption is offset by adjusting tomorrow's consumption until preferences over all later consumption streams coincide: in both, the speed at which an earlier consumption difference fades from habit reveals the decay rate. Our implementation departs from that experiment to make each identifying comparison a single binary choice, which keeps the task intuitive for respondents. Because respondents evaluate their future experience, past spending affects their ranking only through the habit it leaves behind. Varying past rather than future consumption therefore lets a ranking reveal which history leaves more habit (given that habit lowers utility), with no need to search for exactly offsetting consumption, which would require many comparisons. Since actual past spending cannot be changed, the ranking is necessarily hypothetical; it relies on counterfactual reasoning about the past, which respondents handle readily (Section 2.2).

To improve accuracy, the survey uses unfolding brackets to sequentially narrow the range of θ values based on respondents' choices. Each respondent answers a follow-up question that places their θ within one of four predefined brackets (Figure 4). These brackets are designed to ensure that the spending profiles are easily comprehensible to respondents and that they cover an informative range of the parameter space, with the design refined through pilot studies. If a respondent selects Universe One (U1) in the initial question, indicating that their (annualized) θ exceeds 0.5, they are then presented with a follow-up question tied to a threshold of 0.9 (see Appendix Figure A.2 for a screenshot of the question). If they subsequently choose Universe Two (U2), the module concludes, indicating that their habit decay factor lies between 0.5 and 0.9. Table 4 details all thresholds and the corresponding values of ΔC_{U1} and ΔC_{U2} used in the survey. Because the first and last brackets are open-ended, nontrivial mass in boundary brackets is not by itself evidence of bimodality in latent preferences; such mass can also arise from response error. Estimation using the two-wave data helps separate preferences from response error.

Before turning to the statistical model, the raw responses already locate the parameter: in both waves a majority of respondents chose Universe One in the initial habit decay question (54% and 58%; Panel B of Table 1), which by Proposition 1 implies an annual habit decay factor above 0.5. Applying the model (Section 2.5) to the responses yields an estimate of 1.03 for the habit decay rate, corresponding to an (annual) habit decay factor of 0.64 (Table 2). This decay speed implies that nearly 90% of habit depends on spending over the last two years, which is close to findings in the psychology literature indicating that income adaptation takes about two years (for a review, see Clark et al. 2008). The estimate's narrow confidence interval ([0.58, 0.69] at the 95% level), especially relative to the range used in habit-based

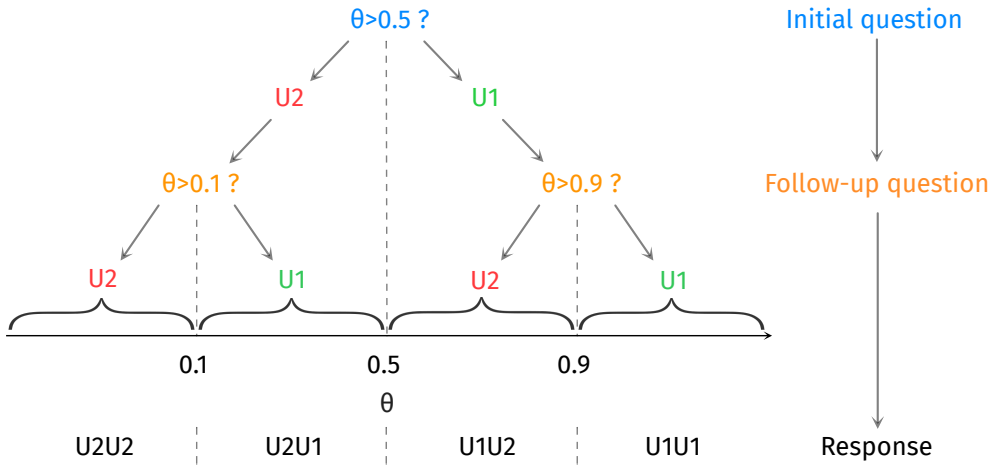


Figure 4: Unfolding Brackets

Notes: U1 and U2: Universe One and Universe Two, respectively. Habit decay rates annualized.

models, makes the decay rate a quantitatively disciplined parameter rather than a free calibration margin. This matters because habit decay can materially affect the quantitative implications of habit formation models: in the canonical model of Campbell and Cochrane (1999), the estimated decay rate implies an expected excess return of 41.1%, against 6.7% in postwar data (Appendix E).

Predicting hedonic adaptation speed. As a validation of our measure of habit decay and our broader methodological approach, we link responses from our survey to responses from a separate survey examining happiness reactions to the 2024 U.S. presidential election (Kimball et al., 2024a).[22] The happiness survey, conducted independently in November and December 2024 on Prolific, included approximately 9,400 respondents. By matching anonymized unique identifiers, we identify 503 of their respondents who also participated in our survey.

The happiness survey measures respondents' happiness both before and after the 2024 U.S. presidential election using a standardized question: "On a scale of 0 to 100, where 0 is the lowest level possible and 100 is the highest level possible, how happy do you feel right now?" Supporters of the election winner tend to report higher happiness, while supporters of the loser tend to report lower happiness post-election, with both groups exhibiting hedonic adaptation toward their pre-election happiness levels (Kimball et al., 2024a). We use the survey's post-election waves to construct a measure of the speed of hedonic adaptation to the election outcome. Specifically, we calculate the absolute daily change in happiness for each respondent between two post-election survey waves at times t_1 and t_2: $\left|\frac{Happiness_{t_2} - Happiness_{t_1}}{t_2 - t_1}\right|$. The median date for t_1 is November 6, 2024, coinciding with the announcement of the election results. To capture the rapid nature of hedonic adaptation to election news, the time difference between t_2 and t_1 is randomized, ranging from 1 to 19 days, with a median of 4 days.

[22] Respondents consented to the potential linkage of their data across multiple surveys.

Table 4: Quantities and Thresholds for Spending Profiles

Question	Quantity and threshold	Baseline = $5,000			Baseline = $3,000		
		Choosing U2	Initial question	Choosing U1	Choosing U2	Initial question	Choosing U1
Habit decay rate	ΔC_{U1}	400	2000	2000	200	1000	1000
	ΔC_{U2}	4000	4000	2200	2000	2000	1100
	Threshold	0.11	0.69	2.40	0.11	0.69	2.40
$-u_H/u_C$	ΔC_{past}	2000	2000	2000	1000	1000	1000
	ΔC_{future}	1000	200	20	500	100	10
	Threshold	0.92	0.55	0.10	0.92	0.55	0.10
$u_{C_{\text{others}}}/u_H$	ΔC_{others}	200	200	1000	100	100	500
	ΔC	4000	2000	2000	2000	1000	1000
	Threshold	1.80	0.90	0.18	1.80	0.90	0.18

Notes: Quantities and implied thresholds mentioned in Propositions 1–3. "Choosing U1" and "Choosing U2": follow-up questions shown after the initial Universe One or Universe Two choice. Quantities are in U.S. dollars.

In the linked dataset, we find a statistically significant correlation (0.13, p-value 0.004) between responses to our survey question on habit decay speed and the hedonic adaptation speed. This suggests that individuals with faster habit decay and thus faster habit adjustment, as measured by our approach, exhibit faster happiness adaptation to the election outcome. Although consumption habit formation and hedonic adaptation are different, their plausible connection and the potential nonlinearity of this relationship render the correlation of 0.13 economically meaningful. Given that both variables are coarse, noisy measures taken two months apart, we interpret this correlation as directional validation—evidence that the elicited parameter tracks real adaptation—rather than as calibrating its magnitude.

To quantify the extent to which individuals with faster habit decay rates adapt hedonically to the election outcome, we regress the measure of hedonic adaptation speed on a binary indicator for habit decay speed exceeding 50% per year. As reported in Column 1 of Table 5, respondents with habit decay speeds above this threshold exhibit, on average, a 1.31-point faster happiness adaptation rate, equivalent to a 40% increase relative to those with slower habit decay (below 50% per year). This finding remains robust when controlling for demographic characteristics and baseline spending levels from our survey, as shown in Column 2 of Table 5.

3.3 Welfare Impacts of Habit Formation and Peer Effects

This subsection details the measurement of two preference parameters expressed as ratios of utility derivatives, both of which are important for resolving the Easterlin paradox (Section 4). The first parameter, $-\frac{u_H}{u_C}$, is the slope of the indifference curve and governs the welfare impact of habit formation relative to one's own consumption. The second parameter, $\frac{u_{C_{\text{others}}}}{u_H}$, assesses the relative welfare impact of peer effects and habit formation. As ratios of utility derivatives, they are naturally identified by trade-offs: habit against one's own consumption for $-\frac{u_H}{u_C}$, and peer effects against habit for $\frac{u_{C_{\text{others}}}}{u_H}$. To identify the

Table 5: Habit Decay Speed Predicts Hedonic Adaptation Speed

	Absolute daily happiness changes	
	(1)	(2)
Fast habit decay (>50%/year)	1.31 (0.48)	1.30 (0.49)
Constant	3.23 (0.35)	5.39 (3.76)
Demographics	N	Y
Baseline spending levels	N	Y
Observations	503	503

Notes: Regression results of absolute daily happiness change on the indicator for annual habit decay speed above 50%. Column 2 adds controls for age, age squared, gender, household income, past and expected monthly household spending, household size, race, education, region, and baseline monthly spending of $3,000. Standard errors in parentheses.

preference parameters while preserving the generality of the utility function, we establish identification up to second-order approximations of the utility differences between choice alternatives, with approximation errors bounded by the remainders of the corresponding Taylor expansions.[23] Exact identification, as established in Proposition 1, entails no approximation.[24]

The slope of the indifference curve, $-\frac{u_H}{u_C}$, reflects the trade-off between habit and (future) spending. To identify it, the survey question uses a trade-off between past spending and future spending (Figure 5). Proposition 2 specifies how such variation identifies $-\frac{u_H}{u_C}$: preferring Universe One over Universe Two means that increasing future spending yields more utility gain than the utility loss from increasing past spending, and therefore u_C is more important than u_H (relative to the threshold $\frac{(\rho+\theta)\Delta C_{\text{future}}}{\rho\Delta C_{\text{past}}+\theta\Delta C_{\text{future}}}$), which explains the direction of the inequality in the proposition.

Proposition 2. *Under a second-order approximation,* $-\frac{u_H}{u_C} < \frac{(\rho+\theta)\Delta C_{future}}{\rho\Delta C_{past}+\theta\Delta C_{future}}$ *if a respondent chooses Universe One over Universe Two for a more satisfying future experience in an indifference-curve slope question.*[25]

As the threshold depends on multiple parameters, the response distribution to this question alone is insufficient to fully characterize this parameter, necessitating the joint estimation of all relevant parameters. The estimated slope of the indifference curve is 0.63 (Table 2), implying that the habit intensity parameter for additive habit (everywhere) and multiplicative habit (in steady state) is about 0.6. This provides direct

[23] The approximation for $-\frac{u_H}{u_C}$ is of second order because the identifying variation is symmetric around the baseline, allowing second-order derivatives of the utility function to cancel out. Similarly, the approximation for $\frac{u_{C_{\text{others}}}}{u_H}$ is of first order, as the corresponding identifying variation lacks this symmetry. The alternative $3,000 spending baseline, which induces smaller spending differences, provides a useful gauge of the magnitude of the approximation errors. Reassuringly, Appendix F.1 shows that parameter estimates are highly robust to the choice of baseline.

[24] The contrast in orders reflects what each parameter asks of the data: the decay rate is recovered from a comparison along a single margin—own habit at fixed future consumption—so monotonicity of u in H pins it down exactly, whereas the welfare parameters are trade-offs between two margins (habit against own consumption for $-u_H/u_C$, peer effects against habit for $u_{C_{\text{others}}}/u_H$) and are therefore identified only up to the order at which those margins are linearized.

[25] In Figure 5, $\Delta C_{\text{past}} = \$2,000$ and $\Delta C_{\text{future}} = \$200$. Table 4 details additional values of ΔC_{past} and ΔC_{future}.



Figure 5: Spending Profiles for a Survey Question on Slope of Indifference Curve

micro evidence for the parameter value often required to match macro data, a magnitude that observational micro-data estimates have frequently failed to yield (Havranek et al., 2017). This estimate indicates that, to a first-order approximation, around 60% of the utility derived from total consumption is eventually habituated. This suggests a substantial welfare impact from habit formation, consistent with Van Praag and Frijters's (1999) finding that approximately 60% of income's effect on happiness is lost over time.[26]

To assess the welfare consequence of peer effects relative to habit formation, we measure $\frac{u_{C_{\text{others}}}}{u_H}$. By accounting for the potential contemporaneous influence of others' spending (i.e., peer effects) in the utility function, $u(C, C_{\text{others}}, H)$, the ratio $\frac{u_{C_{\text{others}}}}{u_H}$ can be elicited by posing a trade-off between others' future spending and one's own past spending. Proposition 3 shows how such variation identifies $\frac{u_{C_{\text{others}}}}{u_H}$. Figure 6 presents the resulting survey question implementing the identifying variation. Preferring Universe One implies that peer effects matter relatively less than habit formation, explaining the direction of the inequality in the proposition.

[26]Fast decay and a large welfare impact are distinct properties. The decay rate governs how quickly habit catches up with consumption, about 90% of the way within two years at the estimated value; the ratio $-\frac{u_H}{u_C}$ governs how much of consumption's marginal utility is offset once habit has caught up. Faster decay therefore brings that offset forward in time without changing its eventual size.



Figure 6: Spending Profiles for a Survey Question on $\frac{u_{C_{\text{others}}}}{u_H}$

Proposition 3. *Under a first-order approximation,* $\frac{u_{C_{others}}}{u_H} < \frac{\rho}{\rho+\theta}\frac{\Delta C}{\Delta C_{others}}$ *if a respondent chooses Universe One over Universe Two for a more satisfying future experience in a* $\frac{u_{C_{others}}}{u_H}$ *question.*[27]

The point estimate for $\frac{u_{C_{\text{others}}}}{u_H}$ is 0.30, suggesting that habit formation has about three times the welfare impact of peer effects.[28] Combining this parameter with the slope of the indifference curve yields an estimate of $\frac{u_{C_{\text{others}}}}{u_C}$ of -0.19. This indicates that a one-dollar increase in others' spending is equivalent to about a 20-cent decrease in one's own spending, which is consistent with existing estimates of the welfare impact of (consumption) peer effects (De Giorgi et al., 2020; Lewbel et al., 2022). Because those estimates rely on observational data and identifying assumptions that this design does not require, their agreement serves as a validation check rather than a restatement of the measurement. Two features of the design bear on interpreting this parameter. First, the survey makes others' spending fully observed, whereas peers' total spending is in practice observed with noise and delay; the estimate is therefore the preference weight on others' consumption conditional on observing it—the application in Section 4 incorporates

[27] In Figure 6, $\Delta C = \$2,000$ and $\Delta C_{\text{others}} = \$200$. Additional values of ΔC and ΔC_{others} are detailed in Table 4.

[28] This paper's model and estimation place no restriction on the sign of $u_{C_{\text{others}}}$, which may be positive (indicating, e.g., altruism) or negative (reflecting, e.g., peer comparison). The estimated sign of $u_{C_{\text{others}}}$ implies that the overall welfare impact of (total-consumption) peer effects is negative.

the imperfect observability through a lag in observing others' consumption. Second, eliciting social preferences in controlled settings is sensitive to scrutiny and framing (Levitt and List, 2007); the survey's non-specific framing and the absence of an experimenter limit both channels, and, though the design cannot rule out bias in either direction, the agreement just noted with field estimates is reassuring.

4 Explaining the Easterlin Paradox

The happiness–income paradox, proposed by Easterlin, states that income and happiness tend to be positively correlated in the short run and in the cross-section, but uncorrelated in the long run (Easterlin, 1973, 1974, 1995, 2001, 2017; Kaiser and Vendrik, 2019; Easterlin and O'Connor, 2025). Alternative views have been proposed, such as that U.S. data might be outliers (Stevenson and Wolfers, 2008), and that life satisfaction can be time-intensive (Kimball and Willis, 2023). Despite the debate, the literature seems broadly in agreement that the empirical gradient of happiness with respect to income is small, and that the cross-section and short-run gradients tend to be larger than the long-run gradient, especially when the time horizon is long enough to cover full economic cycles (Easterlin and O'Connor, 2022). This section explores an explanation of the happiness–income pattern through the lens of habit formation and peer effects. To clarify the intuition behind this explanation, we consider a zero long-run gradient, noting that alternative perspectives can be accommodated with minor adjustments to parameter values without altering the intuition.

Habit formation and peer effects have been the most popular potential explanations for the paradox (Easterlin, 2001; Rayo and Becker, 2007; Clark et al., 2008). Recent evidence on peer effects (Luttmer, 2005; De Giorgi et al., 2020; Lewbel et al., 2022) suggests that they are not powerful enough to fully explain the phenomenon. To our knowledge, evidence on whether habit formation can help explain the paradox is absent from the literature. Based on the measurements above, this section argues that while habit formation alone is also not strong enough to generate the happiness–income pattern of the Easterlin paradox, combining it with peer effects can largely account for it.

Four clarifications merit discussion before proceeding. First, this section focuses on the causal mechanism by which income influences happiness. Everyday life experiences and studies exploiting exogenous variations support this perspective (Frijters et al., 2004; Gardner and Oswald, 2007). Factors affecting happiness other than income offer little help in explaining the paradox because they generally improve with income, making the long-run flat happiness–income gradient even more mysterious (Di Tella and MacCulloch, 2008). Second, this analysis aligns with the Benthamite tradition (Bentham, 1781; Perez-Truglia, 2020), positing that distinctions between happiness and utility exert negligible influence on their relationships with habit formation and peer effects. This assumption is grounded in the complex and ambiguous normative status of their differentiation in contemporary literature (Benjamin et al., 2024, 2014a,b, 2012; Kimball and Willis, 2023). Third, the paradox persists when substituting consumption for income because con-

sumption is closely related to income (Carroll and Summers, 1991),[29] while happiness still has a long-term trend of about zero. Accordingly, the ensuing discussion regards income and consumption as interchangeable. Fourth, the literature identifies three measures of happiness: affect, capturing recent feelings; life satisfaction, reflecting an evaluation of life as a whole; and eudaimonia, encompassing personal growth and meaning. This study focuses on affect and life satisfaction due to their measurement reliability, extensive investigation, and direct relevance to the paradox. Consistent with this focus, Layard and De Neve (2023) argue that eudaimonia is better thought of as a contributor to, rather than a direct measure of, well-being. In the subsequent analysis, affect is proxied by instantaneous utility, and life satisfaction by lifetime utility.

To assess whether habit formation and peer effects can explain the Easterlin paradox, while ensuring robustness to potential misspecification in existing habit formation models (as suggested by the asset-pricing application in Appendix E), we conduct semi-structural simulations grounded in this study's evidence of both phenomena. In particular, the simulations specify agents as influenced by both internal habit formation and peer effects. Habit evolves according to equation (1), with the decay rate calibrated to its estimated value of 1.03. Peer effects arise after agents observe changes in others' consumption, which is assumed to occur k years after those changes.[30]

Under first-order approximations, the effect of habit formation on utility is captured by the ratio $\frac{u_H}{u_C}$, while the impact of peer effects on utility is captured by $\frac{u_{C_{\text{others}}}}{u_C}$. As shown in Section 3.3, the estimates of $\frac{u_H}{u_C}$ and $\frac{u_{C_{\text{others}}}}{u_C}$ both exceed -1 at the 95% confidence level, indicating that habit formation and peer effects, considered separately, are insufficient to eliminate the long-run happiness gain from income growth.

The stylized zero long-run happiness–income gradient requires that

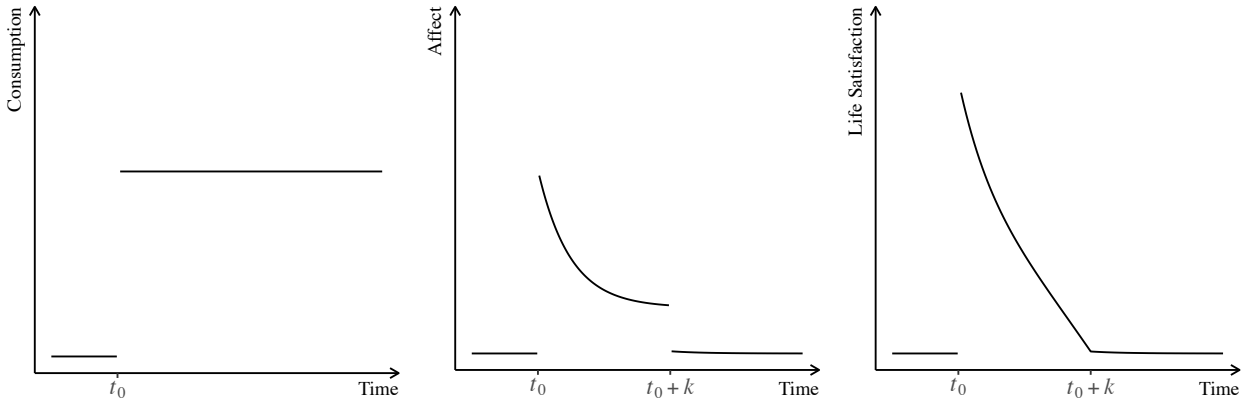
$$\frac{u_H}{u_C} + \frac{u_{C_{\text{others}}}}{u_C} = -1. \tag{2}$$

The point estimate of the left-hand side of the above equation is -0.82: habit formation and peer effects combine to offset 82% of the initial happiness gains from permanent income growth, with a 95% confidence interval spanning a 69% to 96% offset. The estimate is statistically distinguishable from the strict knife-edge of -1 at the 5% level, though the full-offset benchmark remains within the 99% confidence interval; we therefore use the -1 case below as an expository benchmark that transparently illustrates the mechanism, not as an estimate. The estimate of -0.82 leaves a residual gradient equal to just 18% of the short-run effect (95% CI [4%, 31%]). Rather than viewing this as a failure to hit a perfectly flat plateau, this 18% residual aligns with the perspective that habit formation and peer effects are the primary drivers, while leaving room for other proposed explanations—such as inequality, scale use, and economic growth's externalities—to account for the remaining gap.

The intuition of equation (2) is that, to a first-order approximation, habit formation and peer effects fully

[29] Real GDP per capita and real personal consumption expenditure in the United States have a correlation coefficient of 0.99 from 1947 to 2024.

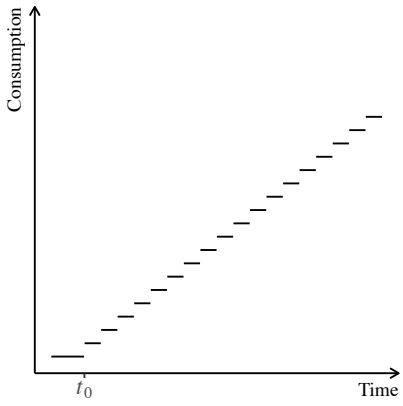
[30] The exact value of k is irrelevant to the explanation, as it only affects the rate of convergence to the steady state.



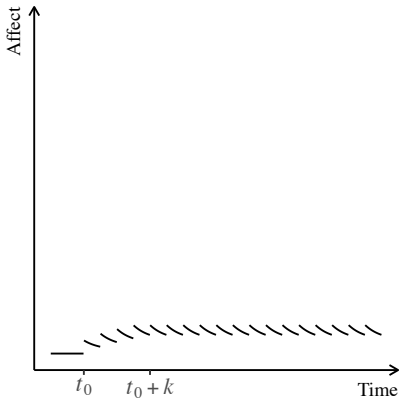
(a) Consumption

(b) Affect

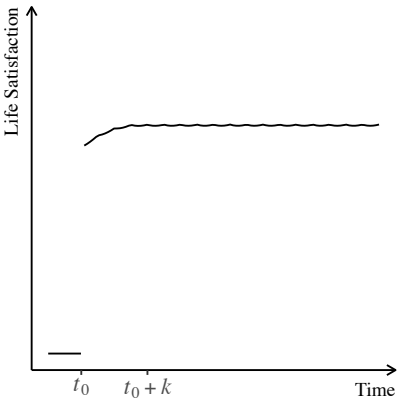
(c) Life satisfaction



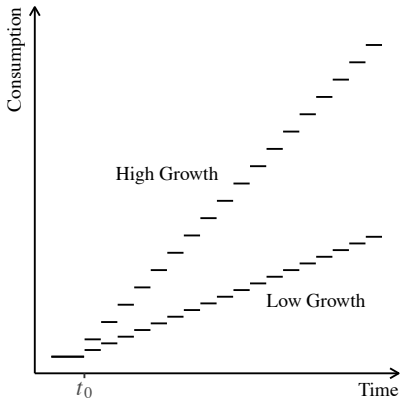
(d) Consumption



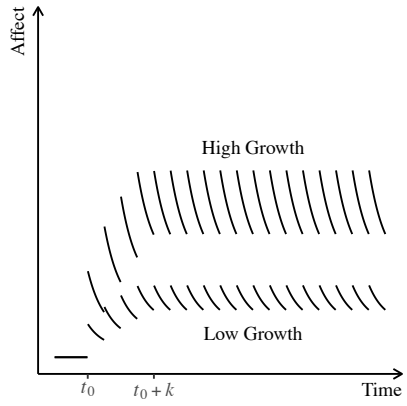
(e) Affect



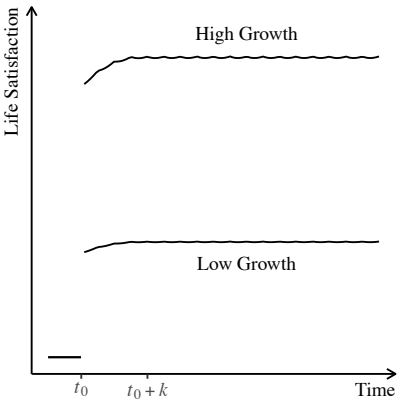
(f) Life satisfaction



(g) Consumption



(h) Affect



(i) Life satisfaction

Figure 7: Happiness Simulations

Notes: Habit decay rate and time discount rate: 1.03 and 0.10, based on the estimates. For intuition, $\frac{u_H}{u_C} = -0.75$ and $\frac{u_{C_{\text{others}}}}{u_C} = -0.25$, preserving measured relative welfare importance and setting their sum to -1. Using exact point estimates (summing to -0.82) would add an upward trend to the plateaus, reflecting the 18% residual gradient discussed above. To make the growth effect on affect visible, Panel (h)'s y-axis is set to 20% of Panel (i)'s. Discontinuities in Panels (e) and (h) arise because consumption is assumed to increase at the start of each year after t_0; continuous changes would smooth them without affecting the analysis.

offset the hedonic impact of permanent changes in consumption over the long run. To see this, consider an economy initially in a steady state, in which residents maintain a constant level of happiness prior to time t_0. Suppose that, from t_0 onward, the economy experiences growth, resulting in a permanent increase in each individual's consumption by a small amount Δc (Figure 7a). Consequently, to the first-order approximation,[31] at t_0, residents' happiness, as measured by affect, rises by $u_C \Delta c$. Over time, however, individuals gradually habituate to this elevated level of consumption, which diminishes affect (Figure 7b). At $t_0 + k$, individuals realize that others in the economy also enjoy the same elevated consumption level. This peer comparison further depresses affect and, in conjunction with habit formation, ultimately restores affect to its initial level. Similarly, life satisfaction exhibits an initial increase followed by a gradual decline, ultimately reverting to its prior steady-state level (Figure 7c). This dynamic, hereafter termed the *wear-off effect*, reflects the combined influence of habit formation and peer effects, which gradually offset the enhancement in happiness induced by a one-episode permanent growth in consumption.

In reality, economies exhibit sustained growth over time, resulting in progressively higher earnings and consumption for individuals. To capture this key aspect, suppose that each agent's consumption rises permanently by Δc at the beginning of every year following t_0, as depicted in Figure 7d. The corresponding evolution of the agent's happiness is illustrated in Figures 7e and 7f. Within each year after t_0, habit formation and peer effects diminish the happiness gains derived from consumption growth, consistent with the dynamics observed in the single-episode growth scenario. However, the long-term trajectory of happiness deviates significantly: rather than steadily reverting to its prior equilibrium, happiness incrementally rises and eventually plateaus at an elevated level. We label these two new patterns of happiness dynamics the *transition effect* and the *plateau effect*.

The transition effect arises from the recurring nature of consumption growth. Each annual increment boosts happiness while initiating a new wear-off effect, both of which stack onto those from previous years. In the short run, this compounding process creates an upward trend in happiness, diverging from the downward trajectory typical of a one-episode growth. Over time, the countervailing forces of habit formation and peer effects progressively mitigate this happiness-enhancing momentum, eventually leading to the plateau effect, where happiness reaches a new steady state.

Because the wear-off effect is proportional to Δc, the transition and plateau effects are also proportional to Δc.[32] Consequently, more rapid consumption growth (Figure 7g) elicits stronger happiness responses (Figures 7h and 7i). This phenomenon, labeled the *growth effect*, implies that higher consumption growth induces more elevated happiness levels during both the transition and plateau phases.

The growth effect explains the positive cross-sectional correlation between income and happiness:

[31] This section focuses on first-order approximations because the identification of $\frac{u_{C_{\text{others}}}}{u_C}$ is under a first-order approximation. This is a good approximation when Δc is small, which is maintained here.

[32] This is a direct implication of first-order approximations. Given positive marginal utility of consumption—a standard assumption in economics—the analysis holds for higher-order approximations: the utility difference between high and low consumption states, though likely reduced, remains positive, preserving qualitative implications.

higher income growth makes people or countries richer and places them on higher happiness trajectories. Economic fluctuations in reality cause consumption to fluctuate, frequently putting the agent into transition phases. The transition effect thus accounts for the short-term positive correlation between income and happiness. Whether income rises or falls, this effect always implies a positive association between income and happiness. The plateau effect explains the zero long-run correlation between income and happiness. Despite sustained income growth, happiness plateaus, yielding a zero correlation between happiness and income in the long run.[33]

An alternative hypothesis for the Easterlin paradox posits that individuals' reporting scales change with income, rather than their actual happiness adapting to income (Deaton, 2008; Prati and Senik, 2024). To control for this rescaling channel, this study's survey employs a binary-choice format and was conducted over a brief period, minimizing the impact and likelihood of potential changes in respondents' scale-use tendencies. To more directly assess rescaling, the survey included visual calibration questions, as introduced by Benjamin et al. (2023), both before and after the core survey modules to measure scale-use consistency. Appendix Figure A.3 shows that respondents' scale-use shifters and stretchers before and after the core modules align closely along the 45-degree line, with an average correlation of 0.84. After adjusting for response errors based on the framework of Benjamin et al. (2023), the correlations indicate essentially perfect consistency, confirming that little rescaling occurred during the survey. This analysis suggests that habit formation and peer effects could explain the Easterlin paradox without rescaling.

How does the above explanation address the questions that motivated the paradox: Does money buy happiness (Easterlin, 1973), and does economic growth improve the human lot (Easterlin, 1974)? To phrase the questions more accurately: if people ultimately care only about happiness and happiness eventually stops increasing with economic growth, should societies continue promoting economic growth after happiness plateaus? The answer implied by the explanation is yes, because happiness would decrease if the economy were to grow at a slower rate. In other words, economic growth initially raises happiness and subsequently maintains it. Slower growth would reduce happiness and stabilize it at a lower level. Maintaining growth therefore remains important even after happiness gains diminish, as it prevents the erosion of well-being that would accompany an economic slowdown.

5 Conclusions

This study introduces a new approach to provide micro-level evidence on habit formation in total-consumption behavior. We specify key parameters governing habit's existence, decay speed, and welfare implications, and formally derive their separate identification through total-consumption variations. By

[33]This time-series plateau is distinct from the cross-sectional happiness–income gradient at high incomes studied by Killingsworth et al. (2023), who find no flattening for most respondents. The two are mutually consistent: in the habit mechanism here, the plateau emerges as habits adjust over time, so at any given moment higher-income individuals—whose habits have not yet caught up—can still report higher happiness.

implementing the variations through survey experiments and validating the results in various ways, we provide evidence on habit formation that is robust to identification challenges in observational, laboratory, or field data. In particular, we find that habit forms internally, decays by two-thirds annually, and generates a welfare effect equivalent to 63% of that from consumption and three times that from peer effects.

Our evidence supports incorporating habit formation into economic models via the standard preference-based approach, with habit formulated relative to one's own past consumption. We show that habit decay can materially affect the performance of habit-based models (Appendix E), and our estimate suggests that this parameter should be disciplined around an annual rate of approximately two-thirds. The micro-level evidence also provides a habit intensity of approximately 0.6, aligning with values commonly used in habit formation models. When peer effects are incorporated, the evidence favors calibrating their welfare impact to roughly one-third that of habit formation.

The measurements also imply that habit formation alone, or peer effects alone, cannot explain the Easterlin paradox, but their combination can account for most of it. People eventually habituate to their income and observe others' similar income changes, largely neutralizing the long-run happiness–income gradient. Before full habituation or realization of others' income changes, happiness moves in the same direction as income changes, explaining the short-run positive happiness–income gradient. Richer people and countries accumulate more happiness changes in the short run, accounting for the positive cross-sectional happiness–income gradient. This mechanism indicates that happiness may rise with increasing income, but only temporarily, before habit formation and peer effects stabilize happiness at a plateau. Although income may lack a long-run effect on happiness, sustained economic growth can remain important for maintaining the stabilized level of happiness. This underscores the important role of policies that promote sustained economic growth, even as the happiness gains from such growth vanish.

These results point to ways in which new measurement approaches can shed light on modern economic theories that are difficult to assess with conventional methods, fostering the parallel development of theories and measurements.

References

Abel, Andrew B, "Asset Prices Under Habit Formation and Catching Up With the Joneses," *American Economic Review*, 1990, *80* (2), 38–42.

Alexander, Samuel, "Living Better on Less? Toward an Economics of Sufficiency," *Simplicity Institute Report 12c*, 2012.

Allcott, Hunt and Todd Rogers, "The Short-Run and Long-Run Effects of Behavioral Interventions: Experimental Evidence from Energy Conservation," *American Economic Review*, 2014, *104* (10), 3003–3037.

_ , Matthew Gentzkow, and Lena Song, "Digital Addiction," *American Economic Review*, 2022, *112* (7), 2424–2463.

Almås, Ingvild, Orazio Attanasio, and Pamela Jervis, "Presidential Address: Economics and Measurement: New Measures to Model Decision Making," *Econometrica*, 2024, *92* (4), 947–978.

Ameriks, John, Andrew Caplin, John Leahy, and Tom Tyler, "Measuring Self-Control Problems," *American Economic Review*, 2007, *97* (3), 966–972.

_ , Joseph Briggs, Andrew Caplin, Matthew D Shapiro, and Christopher Tonetti, "Long-Term-Care Utility and Late-In-Life Saving," *Journal of Political Economy*, 2020, *128* (6), 2375–2451.

Aristotle, *Nicomachean Ethics*, Translated by W. D. Ross. Oxford: Clarendon Press, 1925.

Barsky, Robert B, F Thomas Juster, Miles S Kimball, and Matthew D Shapiro, "Preference Parameters and Behavioral Heterogeneity: An Experimental Approach in the Health and Retirement Study," *The Quarterly Journal of Economics*, 1997, *112* (2), 537–579.

Becker, Gary S. and Kevin M. Murphy, "A Theory of Rational Addiction," *Journal of Political Economy*, 1988, *96* (4), 675–700.

Benjamin, Daniel J, Kristen B Cooper, Ori Heffetz, and Miles S Kimball, "A Well-Being Snapshot in a Changing World," *American Economic Review Papers and Proceedings*, 2019, *109*, 344–49.

_ , _ , _ , and _ , "From Happiness Data to Economic Conclusions," *Annual Review of Economics*, 2024, *16.*

_ , _ , _ , _ , and Jiannan Zhou, "Adjusting for Scale-Use Heterogeneity in Self-Reported Well-Being," *National Bureau of Economic Research*, 2023.

_ , Ori Heffetz, Miles S Kimball, and Alex Rees-Jones, "What Do You Think Would Make You Happier? What Do You Think You Would Choose?," *American Economic Review*, 2012, *102* (5), 2083–2110.

_ , _ , _ , and _ , "Can Marginal Rates of Substitution be Inferred from Happiness Data? Evidence from Residency Choices," *American Economic Review*, 2014, *104* (11), 3498–3528.

_ , _ , _ , and Nichole Szembrot, "Beyond Happiness and Satisfaction: Toward Well-Being Indices Based on Stated Preference," *American Economic Review*, 2014, *104* (9), 2698–2735.

Bentham, Jeremy, "An Introduction to the Principles of Morals and Legislation," *History of Economic Thought Books*, 1781.

Berinsky, Adam J, Gregory A Huber, and Gabriel S Lenz, "Evaluating Online Labor Markets for Experimental Research: Amazon.com's Mechanical Turk," *Political Analysis*, 2012, *20* (3), 351–368.

Block, Henry David and Jacob Marschak, "Random Orderings and Stochastic Theories of Responses," in I. Olkin, S. G. Ghurye, W. Hoeffding, W. G. Madow, and H. B. Mann, eds., *Contributions to Probability and Statistics: Essays in Honor of Harold Hotelling*, Stanford University Press., 1960, pp. 97–132.

Boisseau, Romain P, David Vogel, and Audrey Dussutour, "Habituation in Non-Neural Organisms: Evidence From Slime Moulds," *Proceedings of the Royal Society B: Biological Sciences*, 2016, *283* (1829), 20160446.

Bordalo, Pedro, Katherine Coffman, Nicola Gennaioli, and Andrei Shleifer, "Stereotypes," *The Quarterly Journal of Economics*, 2016, *131* (4), 1753–1794.

Bowman, David, Deborah Minehart, and Matthew Rabin, "Loss Aversion in a Consumption–Savings Model," *Journal of Economic Behavior and Organization*, 1999, *38* (2), 155–178.

Breeden, Douglas T, Robert H Litzenberger, and Tingyan Jia, "Consumption-Based Asset Pricing, Part 2: Habit Formation, Conditional Risks, Long-Run Risks, and Rare Disasters," *Annual Review of Financial Economics*, 2015, *7* (1), 85–131.

Browning, Martin and M. Dolores Collado, "Habits and Heterogeneity in Demands: A Panel Data Analysis," *Journal of Applied Econometrics*, 2007, *22* (3), 625–640.

Brunnermeier, Markus K and Stefan Nagel, "Do Wealth Fluctuations Generate Time-Varying Risk Aversion? Micro-Evidence on Individuals' Asset Allocation," *American Economic Review*, 2008, *98* (3), 713–736.

Bursztyn, Leonardo, Ingar Haaland, Nicolas Roever, and Christopher Roth, "The Social Desirability Atlas," *Journal of Political Economy: Microeconomics*, forthcoming.

Buyalskaya, Anastasia, Hung Ho, Katherine L. Milkman, Xiaomin Li, Angela L. Duckworth, and Colin Camerer, "What Can Machine Learning Teach Us About Habit Formation? Evidence From Exercise and

Hygiene," *Proceedings of the National Academy of Sciences*, 2023, *120* (17), e2216115120.
Camerer, Colin, Yi Xin, and Clarice Zhao, "A Neural Autopilot Theory of Habit: Evidence From Consumer Purchases and Social Media Use," *Journal of the Experimental Analysis of Behavior*, 2024, *121* (1), 108–122.
Campbell, John Y and John H Cochrane, "By Force of Habit: A Consumption-Based Explanation of Aggregate Stock Market Behavior," *Journal of Political Economy*, 1999, *107* (2), 205–251.
Caplin, Andrew, "A Comment on:"Presidential Address: Economics and Measurement: New Measures to Model Decision Making" by Ingvild Almås, Orazio Attanasio, and Pamela Jervis," *Econometrica*, 2024, *92* (4), 979–985.
_ , "Data Engineering for Cognitive Economics," *Journal of Economic Literature*, 2025.
_ **, Mark Dean, and Daniel Martin**, "Search and Satisficing," *American Economic Review*, 2011, *101* (7), 2899–2922.
Carroll, Christopher D and Lawrence H Summers, "Consumption Growth Parallels Income Growth: Some New Evidence," in "National Saving and Economic Performance," University of Chicago Press, 1991, pp. 305–348.
_ **, Jody Overland, and David N Weil**, "Comparison Utility in a Growth Model," *Journal of Economic Growth*, 1997, *2* (4), 339–367.
_ **,** _ **, and** _ , "Saving and Growth with Habit Formation," *American Economic Review*, 2000, pp. 341–355.
Charness, Gary and Uri Gneezy, "Incentives to Exercise," *Econometrica*, 2009, *77* (3), 909–931.
Chen, Xiaohong and Sydney C Ludvigson, "Land of Addicts? An Empirical Investigation of Habit-Based Asset Pricing Models," *Journal of Applied Econometrics*, 2009, *24* (7), 1057–1093.
Chetty, Raj and Adam Szeidl, "Consumption Commitments and Habit Formation," *Econometrica*, 2016, *84* (2), 855–890.
Christiano, Lawrence J, Martin Eichenbaum, and Charles L Evans, "Nominal Rigidities and the Dynamic Effects of a Shock to Monetary Policy," *Journal of Political Economy*, 2005, *113* (1), 1–45.
Chuang, Yating and Laura Schechter, "Stability of Experimental and Survey Measures of Risk, Time, and Social Preferences: A Review and Some New Results," *Journal of Development Economics*, 2015, *117*, 151–170.
Clark, Andrew E, Paul Frijters, and Michael A Shields, "Relative Income, Happiness, and Utility: An Explanation for the Easterlin Paradox and Other Puzzles," *Journal of Economic Literature*, 2008, *46* (1), 95–144.
Cochrane, John H, "Macro-Finance," *Review of Finance*, 2017, *21* (3), 945–985.
Coibion, Olivier, Dimitris Georgarakos, Yuriy Gorodnichenko, Geoff Kenny, and Michael Weber, "The Effect of Macroeconomic Uncertainty on Household Spending," *American Economic Review*, 2024, *114* (3), 645–677.
_ **, Yuriy Gorodnichenko, and Michael Weber**, "Monetary Policy Communications and Their Effects on Household Inflation Expectations," *Journal of Political Economy*, 2022, *130* (6), 1537–1584.
Confucius, *The Analects*, Translated by Simon Leys. W.W. Norton & Company, 1997.
Constantinides, George M, "Habit Formation: A Resolution of the Equity Premium Puzzle," *Journal of Political Economy*, 1990, *98* (3), 519–543.
Crawford, Ian, "Habits Revealed," *The Review of Economic Studies*, 10 2010, *77* (4), 1382–1402.
de Quidt, Jonathan, Johannes Haushofer, and Christopher Roth, "Measuring and Bounding Experimenter Demand," *American Economic Review*, 2018, *108* (11), 3266–3302.
Deaton, Angus, "Income, Health, and Well-Being Around the World: Evidence From the Gallup World Poll," *Journal of Economic Perspectives*, 2008, *22* (2), 53–72.
DellaVigna, Stefano, Attila Lindner, Balázs Reizer, and Johannes F. Schmieder, "Reference-Dependent Job Search: Evidence from Hungary," *The Quarterly Journal of Economics*, May 2017, *132* (4), 1969–2018.
Di Tella, Rafael and Robert MacCulloch, "Gross National Happiness as an Answer to the Easterlin Paradox?," *Journal of Development Economics*, 2008, *86* (1), 22–42.
Diamond, Peter A. and Jerry A. Hausman, "Contingent Valuation: Is Some Number Better Than No Number?," *The Journal of Economic Perspectives*, 1994, *8* (4), 45–64.
Douglas, Benjamin D., Patrick J. Ewell, and Markus Brauer, "Data Quality in Online Human-Subjects Research: Comparisons Between Mturk, Prolific, Cloudresearch, Qualtrics, and Sona," *PLOS ONE*, 03 2023, *18* (3), 1–17.

Duesenberry, James Stemble, *Income, Saving, and the Theory of Consumer Behavior*, Harvard University Press, 1949.

Dynan, Karen E, "Habit Formation in Consumer Preferences: Evidence from Panel Data," *American Economic Review*, 2000, pp. 391–406.

__ **and Enrichetta Ravina**, "Increasing Income Inequality, External Habits, and Self-Reported Happiness," *American Economic Review*, Apr 2007, *97* (2), 226–231.

Easterlin, Richard A, "Does Money Buy Happiness?," *Public Interest*, 1973, *30*, 3.

__ , "Does Economic Growth Improve the Human Lot? Some Empirical Evidence," in "Nations and Households in Economic Growth," Elsevier, 1974, pp. 89–125.

__ , "Will Raising the Incomes of All Increase the Happiness of All?," *Journal of Economic Behavior and Organization*, 1995, *27* (1), 35–47.

__ , "Income and Happiness: Towards a Unified Theory," *The Economic Journal*, 2001, *111* (473), 465–484.

__ , "Paradox Lost?," *Review of Behavioral Economics*, 2017, *4* (4), 311–339.

__ **and Kelsey J O'Connor**, "Explaining Happiness Trends in Europe," *Proceedings of the National Academy of Sciences*, 2022, *119* (37), e2210639119.

__ **and** __ , "The Happiness Revolution in Europe," *Elements in Economics of European Integration*, 2025.

Epstude, Kai and Neal J Roese, "The Functional Theory of Counterfactual Thinking," *Personality and Social Psychology Review*, 2008, *12* (2), 168–192.

Farrell, Diana and Fiona Greig, "Coping with Costs: Big Data on Expense Volatility and Medical Payments," Technical Report, JPMorgan Chase Institute 2017.

Frijters, Paul, John P Haisken-DeNew, and Michael A Shields, "Money Does Matter! Evidence from Increasing Real Income and Life Satisfaction in East Germany Following Reunification," *American Economic Review*, 2004, *94* (3), 730–740.

Fuhrer, Jeffrey C, "Habit Formation in Consumption and Its Implications for Monetary-Policy Models," *American Economic Review*, 2000, pp. 367–390.

Fuster, Andreas, Greg Kaplan, and Basit Zafar, "What Would You Do with $500? Spending Responses to Gains, Losses, News, and Loans," *The Review of Economic Studies*, 2021, *88* (4), 1760–1795.

Gardner, Jonathan and Andrew J Oswald, "Money and Mental Well-being: A Longitudinal Study of Medium-Sized Lottery Wins," *Journal of Health Economics*, 2007, *26* (1), 49–60.

Giorgi, Giacomo De, Anders Frederiksen, and Luigi Pistaferri, "Consumption Network Effects," *The Review of Economic Studies*, 2020, *87* (1), 130–163.

Grishchenko, Olesya V, "Internal vs. External Habit Formation: The Relative Importance for Asset Pricing.," *Journal of Economics and Business*, 2010, *62* (3), 176–194.

Haaland, Ingar, Christopher Roth, Stefanie Stantcheva, and Johannes Wohlfart, "Understanding Economic Behavior Using Open-Ended Survey Data," *Journal of Economic Literature*, December 2025, *63* (4), 1244–80.

Haghani, Milad, Michiel CJ Bliemer, John M Rose, Harmen Oppewal, and Emily Lancsar, "Hypothetical Bias in Stated Choice Experiments: Part I. Macro-Scale Analysis of Literature and Integrative Synthesis of Empirical Evidence From Applied Economics, Experimental Psychology and Neuroimaging," *Journal of Choice Modelling*, 2021, *41*, 100309.

__ **,** __ **,** __ **,** __ **, and** __ , "Hypothetical Bias in Stated Choice Experiments: Part II. Conceptualisation of External Validity, Sources and Explanations of Bias and Effectiveness of Mitigation Methods," *Journal of Choice Modelling*, 2021, *41*, 100322.

Hainmueller, Jens, Dominik Hangartner, and Teppei Yamamoto, "Validating Vignette and Conjoint Survey Experiments Against Real-World Behavior," *Proceedings of the National Academy of Sciences*, 2015, *112* (8), 2395–2400.

Havranek, Tomas, Marek Rusnak, and Anna Sokolova, "Habit Formation in Consumption: A Meta-Analysis," *European Economic Review*, 2017, *95*, 142–167.

Hussam, Reshmaan, Atonu Rabbani, Giovanni Reggiani, and Natalia Rigol, "Rational Habit Formation: Experimental Evidence From Handwashing in India," *American Economic Journal: Applied Economics*, 2022, *14* (1), 1–41.

Jappelli, Tullio and Luigi Pistaferri, "Fiscal Policy and MPC Heterogeneity," *American Economic Journal: Macroeconomics*, 2014, *6* (4), 107–136.

Johnston, Robert J, Kevin J Boyle, Wiktor Adamowicz, Jeff Bennett, Roy Brouwer, Trudy Ann Cameron, W Michael Hanemann, Nick Hanley, Mandy Ryan, Riccardo Scarpa et al., "Contemporary Guidance for Stated Preference Studies," *Journal of the Association of Environmental and Resource Economists*, 2017, *4* (2), 319–405.

Kahneman, Daniel and Amos Tversky, "Prospect Theory: An Analysis of Decision under Risk," *Econometrica*, Mar 1979, *47* (2), 263.

Kaiser, Caspar F. and Maarten C. M. Vendrik, "Different Versions of the Easterlin Paradox: New Evidence for European Countries," *Economics of Happiness*, 2019, pp. 27–55.

Kaplan, Greg and Giovanni L. Violante, "The Marginal Propensity to Consume in Heterogeneous Agent Models," *Annual Review of Economics*, 2022, *14*, 747–775.

Kapteyn, Arie and Federica Teppa, "Hypothetical Intertemporal Consumption Choices," *Economic Journal*, 2003, *113* (486), C140–C152.

Keynes, John Maynard, *The General Theory of Employment, Interest, and Money*, London: Macmillan, 1936.

Khanal, Aditya R., Ashok K. Mishra, and S Nedumaran, "Consumption, Habit Formation, and Savings: Evidence from a Rural Household Panel Survey," *Review of Development Economics*, Sep 2018, *23* (1), 256–274.

Killingsworth, Matthew A., Daniel Kahneman, and Barbara Mellers, "Income and Emotional Well-Being: A Conflict Resolved," *Proceedings of the National Academy of Sciences*, 2023, *120* (10), e2208661120.

Kimball, Miles S and Robert J Willis, "Utility and Happiness," *National Bureau of Economic Research*, 2023.

_ , Claudia R Sahm, and Matthew D Shapiro, "Imputing Risk Tolerance from Survey Responses," *Journal of the American Statistical Association*, 2008, *103* (483), 1028–1038.

_ , _ , and _ , "Risk Preferences in the PSID: Individual Imputations and Family Covariation," *American Economic Review Papers and Proceedings*, Apr 2009, *99* (2), 363–368.

_ , Collin B Raymond, Jiannan Zhou, Junya Zhou, Fumio Ohtake, and Yoshiro Tsutsui, "Happiness Dynamics, Reference Dependence, and Motivated Beliefs in U.S. Presidential Elections," *National Bureau of Economic Research*, 2024.

_ , Daniel Reck, Fudong Zhang, Fumio Ohtake, and Yoshiro Tsutsui, "Diminishing Marginal Utility Revisited," *National Bureau of Economic Research*, 2024.

Kőszegi, Botond and Matthew Rabin, "A Model of Reference-Dependent Preferences," *The Quarterly Journal of Economics*, 2006, *121* (4), 1133–1165.

Kumar, Saten, Yuriy Gorodnichenko, and Olivier Coibion, "The Effect of Macroeconomic Uncertainty on Firm Decisions," *Econometrica*, 2023, *91* (4), 1297–1332.

Kuziemko, Ilyana, Michael I Norton, Emmanuel Saez, and Stefanie Stantcheva, "How Elastic Are Preferences for Redistribution? Evidence from Randomized Survey Experiments," *American Economic Review*, 2015, *105* (4), 1478–1508.

Larrimore, Jeff, Alicia Lloro, Zofsha Merchant, Ellen A Merry, Fatimah Shaalan, Julie Siwicki, and Mike Zabek, "Economic Well-Being of US Households in 2024," Technical Report, Board of Governors of the Federal Reserve System (US) 2025.

Layard, Richard and Jan-Emmanuel De Neve, *Wellbeing*, Cambridge University Press, 2023.

Levitt, Steven D. and John A. List, "What Do Laboratory Experiments Measuring Social Preferences Reveal About the Real World?," *Journal of Economic Perspectives*, 2007, *21* (2), 153–174.

Lewbel, Arthur, Samuel Norris, Krishna Pendakur, and Xi Qu, "Consumption Peer Effects and Utility Needs in India," *Quantitative Economics*, 2022, *13* (3), 1257–1295.

List, John A, "Do Explicit Warnings Eliminate the Hypothetical Bias in Elicitation Procedures? Evidence From Field Auctions for Sportscards," *American Economic Review*, 2001, *91* (5), 1498–1507.

Liu, Xuemei, Juan Lai, Chuanliang Han, Hao Zhong, Kang Huang, Yuanming Liu, Xutao Zhu, Pengfei Wei, Liming Tan, Fuqiang Xu, and Liping Wang, "Neural Circuit Underlying Individual Differences in Visual Escape Habituation," *Neuron*, 2025.

Ljungqvist, Lars and Harald Uhlig, "Tax Policy and Aggregate Demand Management Under Catching up With the Joneses," *American Economic Review*, Jun 2000, *90* (3), 356–366.

Loewenstein, George and Drazen Prelec, "Preferences for Sequences of Outcomes," *Psychological Review*, 1993, *100* (1), 91–108.

__ **and Nachum Sicherman**, "Do Workers Prefer Increasing Wage Profiles?," *Journal of Labor Economics*, 1991, *9* (1), 67–84.

__ **, Ted O'Donoghue, and Matthew Rabin**, "Projection Bias in Predicting Future Utility," *The Quarterly Journal of Economics*, 2003, *118* (4), 1209–1248.

Lucas, Robert E, "Econometric Policy Evaluation: A Critique," in "Carnegie-Rochester Conference Series on Public Policy," Vol. 1 1976, pp. 19–46.

Luttmer, Erzo F. P., "Neighbors as Negatives: Relative Earnings and Well-Being," *The Quarterly Journal of Economics*, 08 2005, *120* (3), 963–1002.

Manski, Charles F, "Measuring Expectations," *Econometrica*, 2004, *72* (5), 1329–1376.

Markowitz, Harry, "The Utility of Wealth," *Journal of Political Economy*, 1952, *60* (2), 151–158.

Masatlioglu, Yusufcan and Efe A Ok, "A Canonical Model of Choice With Initial Endowments," *The Review of Economic Studies*, 2014, *81* (2), 851–883.

Matousek, Jindrich, Tomas Havranek, and Zuzana Irsova, "Individual Discount Rates: A Meta-Analysis of Experimental Evidence," *Experimental Economics*, 2022, *25* (1), 318–358.

Matyskova, Ludmila, Brian Rogers, Jakub Steiner, and Keh-Kuan Sun, "Habits as Adaptations: An Experimental Study," *Games and Economic Behavior*, 2020, *122*, 391–406.

Meghir, Costas and Guglielmo Weber, "Intertemporal Nonseparability or Borrowing Restrictions? A Disaggregate Analysis Using a U.S. Consumption Panel," *Econometrica*, Sep 1996, *64* (5), 1151.

Mullainathan, Sendhil, "A Memory-Based Model of Bounded Rationality," *The Quarterly Journal of Economics*, 08 2002, *117* (3), 735–774.

Naik, Narayan Y and Michael J Moore, "Habit Formation and Intertemporal Substitution in Individual Food Consumption," *Review of Economics and Statistics*, 1996, *78* (2), 321–328.

Peer, Eyal, David Rothschild, Andrew Gordon, Zak Evernden, and Ekaterina Damer, "Data Quality of Platforms and Panels for Online Behavioral Research," *Behavior Research Methods*, 2022, *54* (4), 1643–1662.

Perez-Truglia, Ricardo, "The Effects of Income Transparency on Well-Being: Evidence from a Natural Experiment," *American Economic Review*, April 2020, *110* (4), 1019–54.

Pinsker, Harold, Irving Kupfermann, Vincent Castellucci, and Eiuc Kandel, "Habituation and Dishabituation of the Gill-Withdrawal Reflex in Aplysia," *Science*, 1970, *167* (3926), 1740–1742.

Pollak, Robert A, "Habit Formation and Dynamic Demand Functions," *Journal of Political Economy*, 1970, *78* (4), 745–763.

Post, Thierry, Martijn J Van den Assem, Guido Baltussen, and Richard H Thaler, "Deal or No Deal? Decision Making Under Risk in a Large-Payoff Game Show," *American Economic Review*, 2008, *98* (1), 38–71.

Praag, Bernard M. S. Van and Paul Frijters, "The Measurement of Welfare and Well-Being: The Leyden Approach," in "Well-Being: Foundations of Hedonic Psychology," Russell Sage Foundation, 1999, pp. 413–433.

Prati, Alberto and Claudia Senik, "Is It Possible to Raise National Happiness?," *HAL SHS Working Paper 04850502*, 2024.

Ravn, Morten, Stephanie Schmitt-Grohé, and Martin Uribe, "Deep Habits," *The Review of Economic Studies*, 2006, *73* (1), 195–218.

Rayo, Luis and Gary S Becker, "Evolutionary Efficiency and Happiness," *Journal of Political Economy*, 2007, *115* (2), 302–337.

Roth, Christopher and Johannes Wohlfart, "How Do Expectations about the Macroeconomy Affect Personal Expectations and Behavior?," *The Review of Economics and Statistics*, 10 2020, *102* (4), 731–748.

Royer, Heather, Mark Stehr, and Justin Sydnor, "Incentives, Commitments, and Habit Formation in Exercise: Evidence From a Field Experiment With Workers at a Fortune-500 Company," *American Economic Journal: Applied Economics*, 2015, *7* (3), 51–84.

Rozen, Kareen, "Foundations of Intrinsic Habit Formation," *Econometrica*, 2010, *78* (4), 1341–1373.

Ryder, Harl E and Geoffrey M Heal, "Optimal Growth With Intertemporally Dependent Preferences," *The Review of Economic Studies*, 1973, *40* (1), 1–31.

Sergeyev, Dmitriy, Chen Lian, and Yuriy Gorodnichenko, "The Economics of Financial Stress," *The Review of Economic Studies*, 2024, p. rdae110.

Shi, Shouyong and Larry G. Epstein, "Habits and Time Preference," *International Economic Review*, 1993, *34* (1), 61–84.

Smets, Frank and Rafael Wouters, "Shocks and Frictions in US Business Cycles: A Bayesian DSGE Approach," *American Economic Review*, 2007, *97* (3), 586–606.

Smith, Adam, *An Inquiry into the Nature and Causes of the Wealth of Nations*, London: W. Strahan and T. Cadell, 1776.

Spinnewyn, Frans, "Rational Habit Formation," *European Economic Review*, 1981, *15* (1), 91 – 109.

Stantcheva, Stefanie, "How to Run Surveys: A Guide to Creating Your Own Identifying Variation and Revealing the Invisible," *Annual Review of Economics*, 2023, *15* (1), 205–234.

Stevenson, Betsey and Justin Wolfers, "Economic Growth and Subjective Well-Being: Reassessing the Easterlin Paradox," *Brookings Papers on Economic Activity*, 2008, *2008* (1), 1–87.

Stiglitz, Joseph E, "Where Modern Macroeconomics Went Wrong," *Oxford Review of Economic Policy*, 2018, *34* (1-2), 70–106.

Strauss, Milton E and Gregory T Smith, "Construct Validity: Advances in Theory and Methodology," *Annual Review of Clinical Psychology*, 2009, *5*, 1–25.

Thakral, Neil and Linh T. Tô, "Daily Labor Supply and Adaptive Reference Points," *American Economic Review*, 2021, *111* (8), 2417–2443.

Tserenjigmid, Gerelt, "On the Characterization of Linear Habit Formation," *Economic Theory*, 2020, *70* (1), 49–93.

Tversky, Amos and Daniel Kahneman, "Loss Aversion in Riskless Choice: A Reference-Dependent Model," *The Quarterly Journal of Economics*, 1991, *106* (4), 1039–1061.

Walker, W. Richard, Rodney J. Vogl, and Charles P. Thompson, "Autobiographical Memory: Unpleasantness Fades Faster than Pleasantness over Time," *Applied Cognitive Psychology*, 1997, *11* (5), 399–413.

Webb, Ryan, Jessica Fong, Peter Landry, Julia Levine, Alexendra Steiny Wellsjo, Olivia Natan, Asaf Mazar, Clarice Y. Zhao, Phillippa Lally, Sanne de Wit, John P. O'Doherty, Andrew T. Ching, Raphael Thomadsen, Matthew Osborne, Mark E. Bouton, Wendy Wood, and Colin F. Camerer, "Integrating Neuro-Psychological Habit Research Into Consumer Choice Models," *International Journal of Research in Marketing*, 2026, *43* (2, Part B), 514–536.

Weidmann, Ben and David J Deming, "Team Players: How Social Skills Improve Team Performance," *Econometrica*, 2021, *89* (6), 2637–2657.

Wiswall, Matthew and Basit Zafar, "Preference for the Workplace, Investment in Human Capital, and Gender," *The Quarterly Journal of Economics*, 2018, *133* (1), 457–507.

Wood, Wendy and Dennis Rünger, "Psychology of Habit," *Annual Review of Psychology*, 2016, *67* (1), 289–314.